



GOLDEN
TREE

Wonderworld

A Book of Environmental Studies

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Class
5





NEP 2020

The **National Education Policy** (NEP) for India was approved by the **Union Cabinet of India** on 24 July 2020. It is based on four pillars which are Access, Equity, Quality and Accountability.

The main aim of the NEP is to improve the performance of students through various criteria. The main features of the NEP can be understood under the following heads:

Creativity

Students will be given some kind of activity to related to the chapter to make clear the vision of the chapter. It will help to enhance imagination.



Art Integration

This will relate the chapter with some art activity.



Experimental Learning

Through this, students will learn by doing. It is like DIY or Do it yourself.



Life Skills & Values

It will help to inculcate the moral values in the students.





Subject Enrichment

It will help to understand the chapter more through related subject matter.



Critical Thinking

It will help the students how to act or behave in any situation.



Computational Thinking

It will enhance the problem solving capability of the students.



Inter Disciplinary

It will relate the subject matter with other aspects.



Communication

Through this, the students will be encouraged to ask or to know more.



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FORWARD

The series '**WONDERWORLD**' on Environmental Studies is based on the guidelines laid down by the CBSE Board. **New Education Policy (NEP)** focuses on the need of Environmental Studies. As the world is progressing in its environmental problems such as climate change, global warming, deforestation and pollution the need of environmental education is necessary. In New Education Policy, a sustainable development has been given on Environmental Education. The topics that have been included in this book make the transition from primary to middle school easier for the learners.

The main features of this series are as follows:

1. Powerful visual appeal with multi-coloured pictures.
2. Enjoyable fun activities.
3. Interactive and child centered approach has been followed to arouse interest in the students to enhance their level of curiosity.

We hope that parents and teachers will give prudential consideration and realistic approach to the book. Any suggestions for the improvement for this book shall be gratefully acknowledged and incorporated in the subsequent editions.

— Author & Publisher



Cleanliness of the Body Parts

Health is the most valuable thing for everybody because without health we cannot live happily.

Indeed 'Health is Wealth'. An unhealthy person cannot enjoy his life and cannot do anything properly.

To being healthy, we should follow some rules :



Rest And Sleeping Time

Rest is must for good health. Our daily routine should be fixed.

We should go to bed and sleep around 9:00 p.m. or 10:00 p.m. in the night.

We should leave the bed early in the morning. It is true that "Early to bed and early to rise, makes a man healthy, wealthy and wise."



Care Of Teeth

We should rinse our mouth when we get up in the morning. We should brush our



teeth regularly twice a day to keep our teeth and gums healthy. We should brush our teeth in a proper way. We should not brush very forcibly; if we do so, then the protective layer of teeth (enamel) will destroy. We should rinse our mouth after eating anything because very small particles of food get stuck in between the teeth, and bacteria can grow in the mouth. After sometime, bad smell starts coming and teeth become yellow.



We should eat fibrous fruits and vegetables. They help us to keep our gums healthy. Rub your gums well with fingers. If our gums are unhealthy, our teeth can never be strong.



We should take a complete bath daily. Dust particles and various disease causing germs stick to the skin. They may harm our skin if we do not bath regularly. We should use a good quality soap. Armpits, thighs, earlobes, eyelids etc. should be clean properly.

We should wash our hair properly with soap or shampoo. We should wash our hair properly with water.



After bathing, we should dry our body with clean towel.

We should oil and comb our hair daily.



After playing games or in many other activities, our hands and feet become dirty. Many germs can enter in our body if we do not wash them properly.

We should wash our hands and feet with antiseptic soap or lotion. After applying soap, we should rub our hands properly and then remove the soap with water. We should wash our hands before eating anything.





Care Of Sense Organs



Sense organs are very delicate organs so we should take care of them very carefully.

Eyes are very important and most sensitive part of our body. We should wash our eyes with clean and cold water many times in a day.

We should not rub our eyes with dirty hands or clothes. It may cause infection.

Ears are also very delicate organs of the body. We should not clean our ears with matchstick or pointed things. We should only use ear bud. We should not pour oil in our ears.



Nose is also a sense organ. We must not insert our finger or pointed thing in the nose. It may harm us. We should clean our nose daily with a clean cloth.

Tongue is an important organ of our body. It helps us to taste different things. We should clean our tongue daily while brushing.



Skin is very sensitive organ of our body. Dust particles and disease causing germs settle on our body. We should clean our skin with antiseptic lotion or soap.



Regular Exercise

Regular exercise keeps us fit and healthy. Exercise helps to make our muscles and bones strong.



Regular exercise makes our body active. Walking, running, jogging, cycling, yoga, swimming etc. are some exercises.



REWIND

- ▶ For being healthy, our daily routine should be fixed, i.e. sleeping and rising time.
- ▶ We should brush our teeth regularly. We should rinse our mouth after eating anything.
- ▶ We should take bath daily with soap.
- ▶ We should take proper care of our eyes, ears, nose, tongue and skin.
- ▶ Regular exercise keeps us fit and healthy.



A. Multiple choice questions.

Tick (✓) the correct option :

- We should brush our _____.
(a) teeth ☐ (b) gums ☐ (c) hair ☐
- We should wash our _____ before eating.
(a) feet ☐ (b) mouth ☐ (c) hands ☐
- Eyes should be washed with _____.
(a) cold water ☐ (b) warm water ☐ (c) boiled water ☐
- Exercise keeps our body _____.
(a) fit ☐ (b) healthy ☐ (c) both (a) and (b) ☐



B. Write 'T' for true and 'F' for false :

1. We should not bath daily.
2. We should brush our teeth twice a day.
3. We should oil and comb our hair daily.
4. We should clean our ears with pin.



C. Match the column A with column B :

- | A | | B |
|----------|-------|--------------|
| 1. Teeth | (i) | oil and comb |
| 2. Hair | (ii) | sensitive |
| 3. Skin | (iii) | exercise |
| 4. Yoga | (iv) | cold water |
| 5. Eyes | (v) | brush |

D. Answer the following questions :

1. How should we take care of our teeth?

2. How should we clean our body parts while bathing?

3. Why should we wash our hands and feet after playing?

4. How should we take care of our eyes?

5. How should we take care of our ears?

Extra Curriculum



- Make a chart to show that how should we take care of our sense organs.





Types of Houses

House is a place where we can keep our belongings, feel comfortable and live with our family members.



Types Of Houses

There are many reasons why people build different types of houses.



Climatic Conditions

Hot places

In the places where the climate is hot, houses are generally built of **bricks** and **stones**.

Such houses have **thick walls** and **flat rooftops**. Thick walls keep the inside of the house cool. Flat rooftops can be used as a terrace.

They can be used for drying clothes and for sleeping.



Cold places

In places that receive heavy snowfall, people build houses with **sloping rooftops**.

This type of rooftop lets the snow slide down easily. This does not allow the house to become damp and cold.



Heavy rainfall receiving areas

In some places, it rains so much that the land keeps getting flooded.

Houses in such places are often built on sets of wooden posts called **stilts**.

These houses are quite high above the ground.

Such houses are safe from floods and dangerous animals such as snakes and insects. A ladder is used to enter the house. At night, the ladder is removed from the house for safety.

Such houses also have sloping roofs to let the rainwater fall down easily.



Houseboat

A houseboat is a boat with a house on it. It floats on a lake.

Dal lake in **Srinagar** is internationally famous for its houseboats.



Igloo

An igloo is a house made of snow. It is dome-shaped and has a small door.

Igloos are found in polar regions, places covered with snow throughout the year.

Eskimos live in igloos.



In hilly, forested areas, people use a lot of wood to build houses, as wood in such places is easily available. In places where stones are easily available, people use blocks of stone to make their houses.



Tent

Tents are used as temporary houses by mountaineers, campers, soldiers and scouts. They can be carried anywhere. Tents are made of **canvas**.



Caravan

A caravan is a movable house. It is pulled by a car engine or animals.

It is designed for people to live and sleep in.

There are various reasons, why different types of houses are made in the same place.



Cost Of Construction

The cost of making a house is another reason why houses in the same place are different.

Chawl

A **chawl** consists of rows of one-room houses. The cost of constructing houses in a chawl is not very high.

Chawls are built in densely populated areas where not much space is available.



Kachcha house

It does not take much money to make **kachcha houses**. Such houses are built with bamboo, clay and straw. A kachcha house with a roof of straw is called a **thatched hut**. Kachcha houses are not very strong.



Pucca houses

Pucca houses are made from building materials such as bricks, cement, steel and concrete. Bungalows, flats and multi-storeyed buildings are pucca houses or permanent houses.

You need a lot of money to construct a pucca house.





- ▶ People build different types of houses because of climatic conditions, landforms, available materials and needs.
- ▶ Different types of houses are made in the same place because of the cost of construction and so on.



A. Multiple choice questions.

Tick (✓) the correct option :

1. We find houses with sloping roofs in _____.
 (a) plains (b) mountains
 (c) deserts (d) none of these
2. A kachcha house with a roof of straw is called a _____.
 (a) chawl (b) bungalow
 (c) thatched hut (d) tent
3. An igloo is made up of _____.
 (a) snow (b) wood
 (c) canvas (d) none of these
4. A shortage of space has resulted in the construction of a _____.
 (a) kachcha house (b) chawl
 (c) multi-storeyed building (d) none of these

B. Fill in the blanks :

1. Houses built on wooden posts are called _____.
2. _____ is a movable house.
3. An _____ is a house made up of snow.
4. A _____ is a house made up of canvas.



C. Answer the following questions :

1. Why do we need a house?

2. Why do people built stilt houses in water-flooded areas?

3. Name the materials with which multi-storeyed buildings are constructed.

4. List four reasons why people build different types of houses.

5. Why are wood houses made in hilly, forested areas?

Extra Curriculum



■ **Write the names of the following pictures.**



Subject Enrichment

■ **Make a different kind of house using straw and matchsticks.**

■ **Paste the picture of your house in the space provided.**

Creativity

Art Integration





Eating Together



Eating Together In A Family

Eating together in a family creates a bond of oneness among the members of the family. It brings family members closer to one another. It is while eating together, that we share our experiences, ideas and thoughts with our family members.



Eating Together In School



Now-a-days, most of the government schools in our country provide **midday meals** to students. As you know, many children in our country do not get even one full meal a day. These children cannot study properly because they do not get a proper meal. A few years ago, the **Supreme Court** gave the order that all government run primary schools must provide their students with hot cooked food.



Before the midday meal starts, the teacher sees that the children wash their hands properly. Then they sit in an orderly manner and wait for the food to be served. A variety of food is served. Sometimes students have vegetable and roti or aloopuri, sometimes dal-rice and sometimes idlis. Everyday the teacher ensures that children change their place. This gives the children the opportunity to make new friends.

A menu plan is displayed on the notice board outside the Principal's office. It tells the children the kind of food they will get on each day of the week. Special dishes are served on important days like **Republic Day** and **Diwali**.



Eating Together On Special Occasions

We usually get together with our relatives and friends on special occasions like festivals, weddings, and birthdays. Special dishes are cooked on these occasions. We engage special cooks called **halwais** to make special food for us. They prepare food in large vessels. The food is placed in serving bowls and served in good crockery. Eating together on such occasions brings a lot of joy and happiness to all of us. It creates a feeling of oneness among our family members, relatives and friends.



Eating Together In Boarding Schools



Some students study in schools where they live as well. Such schools are called **boarding schools**. In a boarding school, students sit together in a large dining hall called **hostel mess** to have their meal. They have fixed time for their meals. The hostel warden and house masters supervise to ensure that the students eat properly.





Eating Together In A Community

When a large number of people eat together at a special place, the activity is called **community eating**. People often gather together in **gurudwaras** to eat the same food. Food is prepared free of charge. People sit together on the floor and have the same food at a time. The **langar/prasad** consists of **dal**, **roti**, **sabzi** and **salad**. It is prepared in the common kitchen and serving the food to people is called as **seva** or service to mankind.



Eating Together During Lunch Break



We have a lot of fun during lunch break at our school. We all share our meals during lunch time. There is a wide variety of dishes to choose from! Sharing our meals also brings us closer to our friends.

REWIND



- ▶ Eating together in a family creates a bond of unity.
- ▶ In boarding schools, there is a fixed time for meals.
- ▶ People often gather together in gurudwaras to eat the langar/prasad.
- ▶ Eating together on occasions brings a lot of joy and happiness.
- ▶ Sharing our meals brings us closer to our friends.





A. Multiple choice questions.

Tick (✓) the correct option :

1. The food served in a government run school during the day is called _____ meal.
(a) morning ☐ (b) midday ☐
(c) afternoon ☐ (d) dinner ☐
2. Schools where students study and live are called _____.
(a) day schools ☐ (b) daylong schools ☐
(c) boarding schools ☐ (d) none of these ☐
3. We can get langar/prasad in a _____.
(a) gurudwara ☐ (b) church ☐
(c) hospital ☐ (d) school ☐
4. In a boarding school, students have _____ time for their meal.
(a) any ☐ (b) no ☐
(c) fixed ☐ (d) none of these ☐

B. Fill in the blanks :

1. We share our experiences, thoughts and _____ with our family members.
2. Eating together creates a bond of _____ among members.
3. Langar is prepared in a large _____.
4. Boarding schools have large dining halls called _____.

C. Answer the following questions :

1. Why should family members eat together?

2. Name two occasions when we eat together in large groups.

3. How is food served in a boarding school?



4. Why should we share our meals at schools?

5. What is the midday meal scheme? Why is it important?

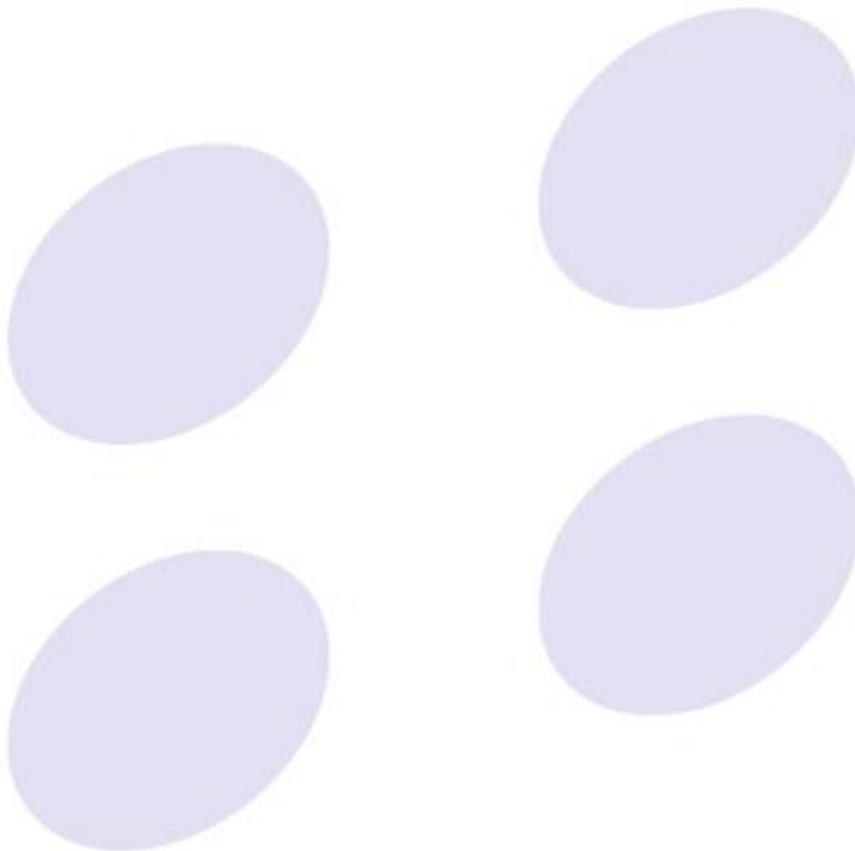
Extra Curriculum



- Look at the following pictures and also write their names in the boxes. **Subject Enrichment**



- Collect pictures of your family members and friends meeting on special occasions like festivals, weddings and birthdays and paste them in the space provided. **Creativity**





Breathe In Breathe Out



We breathe air to live. Breathing is the taking in and throwing out air. Two big spongy organs called lungs help us in this process. Our lungs are encased in a **rib cage** made of bones.

The lungs expand when we breathe in and contract when we breathe out. The process of breathing in air is called **inhalation** or **inspiration**. The process of breathing out air is called **exhalation** or **expiration**. We inhale a gas called **oxygen** and exhale a gas called **carbon dioxide**. This whole process is called **respiration**.

The respiratory system in our body helps us to carry out the process of respiration. The main organs of the respiratory system are the **nose**, **windpipe** and **lungs**. We breathe in air through our nose. Our lungs are connected to nose through the **windpipe**.

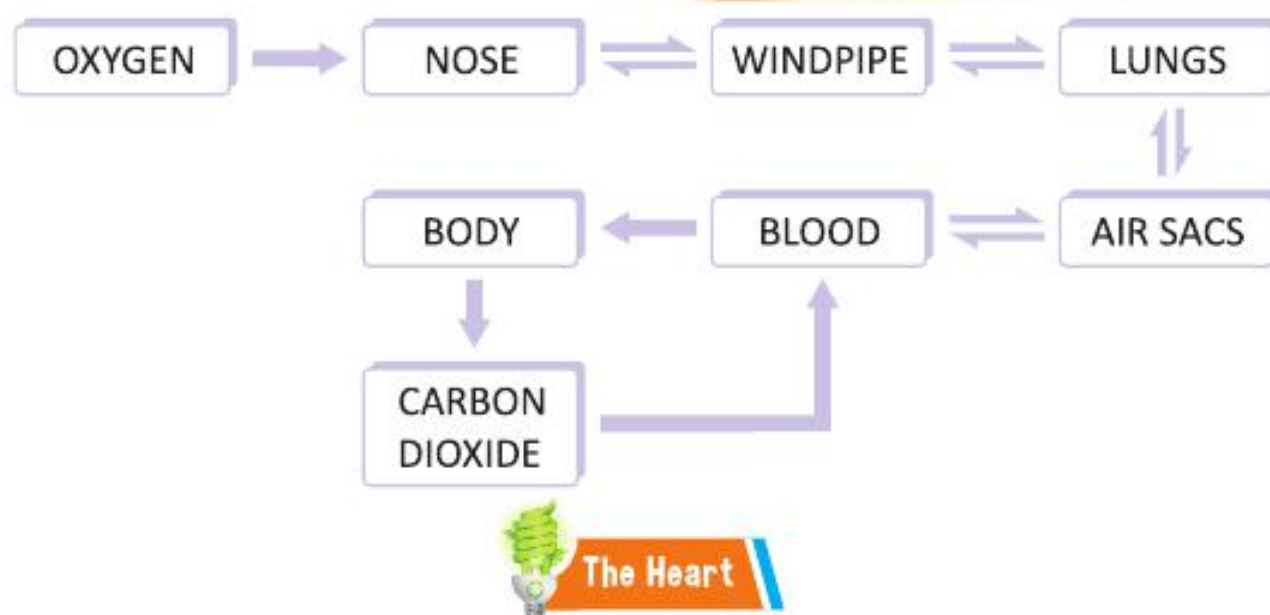
There are millions of **air sacs** inside the lungs. Air sacs are covered with a network of tubes that carry blood around the body. Oxygen, which is inhaled, comes to the air sacs through the windpipe and get absorbed by the blood. The blood carries the oxygen to the whole body. Along with the blood the nutrients also come to make energy.

Carbon dioxide is released in this process. This gas is harmful for the body. That is why, it is sent through the blood to the lungs. During exhalation, carbon dioxide comes out of our body.



The small hairs inside the nose prevent dust particles from entering our body when we breathe in through the nose. But, if we breathe in through the mouth, dust particles can easily go inside us and cause infection.

Study the flowchart given below and note the direct process of taking in oxygen and the reverse path taken by carbon dioxide.



BOTTOM LINE

Do not hold your breath! You have probably noticed that if you breathe in and hold your breath for long, you start to feel tired. This is because your body is running out of oxygen. It is dangerous to hold your breath for too long. You may become unconscious if you do too often.

Have you ever accidentally cut your finger or have you even fallen down and hurt your knees and elbows? What happened then?

You must have seen blood oozing out of the wound.

Blood is very special liquid. We also call it the **river of life**. An average adult human body has 4 to 5 litres of blood. It circulates all over the body in 2 to 3 minutes. The journey of blood around the body is called **blood circulation**.

You must be wondering how blood moves through various parts of our body. There is a pumping station that helps our blood travel through the body. This pumping station is our **heart**. The heart beats approximately 100,000 times a day to perform this

BOTTOM LINE

Doctors listen to the heartbeat of patients by using a special instrument called a stethoscope.



function. A resting heart may pump blood or beat as slowly as 60 times in a minute. But, when we exercise, our heart beats 100 times in a minute or more.



Be Your Own Doctor

We can feel the pumping action of the heart by a simple method. We can feel the heartbeat when we touch the back of the wrist of a person. Place the fingers of your right hand on the back of your left wrist, a little below your thumb. You can feel the **pulse** or the regular movement of your blood through your body. Now, count your pulse rate or the number of times your heart beats in a minute.



Our heart muscle keeps on beating from the moment we are born until the moment we die. It never tires.



Blowing Hot And Blowing Cold

We use our breath in many ways. We blow on something hot to cool it. We blow on hot milk or tea before drinking it so that it does not burn our lips. We also blow on our hands when we feel cold to make them warmer.

Open your mouth wide and blow hard on your hands. You will feel warm and humid air coming out from your mouth. That is because our body has a steady warm temperature inside.



Temperature of the air is how hot or cold it is. Now, open your mouth as if you can whistle and blow air on your hands. You will feel cool air on your hands.

Have you seen someone cooking food on a **Chulha or coal stove**? They use coal and wood as fuel and frequently blow air through a pipe to light a fire. This is done because even fire needs oxygen to burn. When we blow on fire, the ash particles fly aside.

Thus, the burning things get more oxygen and start burning faster.





- ▶ The process of taking in oxygen is called inhalation.
- ▶ The process of giving out carbon dioxide is called exhalation.
- ▶ The process of inhalation and exhalation is called respiration.
- ▶ The heart pumps blood all over the body.
- ▶ The main organs of the respiratory system are the nose, windpipe and the lungs.
- ▶ The journey of blood around the body is called circulation.



A. Multiple choice questions.

Tick (✓) the correct option :

1. The process of taking in oxygen through the nose is known as _____.
 (a) inhalation ☐ (b) exhalation ☐ (c) respiration ☐
2. _____ is known as the river of life.
 (a) pulse ☐ (b) blood ☐ (c) fuel ☐
3. The spongy organs that help us in respiration are _____.
 (a) lungs ☐ (b) heart ☐ (c) nose ☐

B. Write 'T' for true and 'F' for false :

1. It is better to breathe through our mouth than our nose. ☐
2. The rib cage is made of bones. ☐
3. Lungs are not connected to the nose. ☐
4. We exhale oxygen. ☐
5. Our heart beats faster when we do exercise. ☐



C. Match the column A with column B :

A

1. Inhalation
2. Blowing over hot food
3. Beating heart
4. Inhaling and exhaling
5. Wrist

B

- (i) respiration
- (ii) pumping blood
- (iii) oxygen
- (iv) pulse rate
- (v) cold air

D. Answer the following questions :

1. What is respiration?

2. Which organs of the body help us in respiration?

3. What is blood circulation?

4. What does the heart do?

5. How can we check our heart beat?

Extra Curriculum



- Find six things related to respiration in this word grid.

S	T	K	M	I	S	O	T	N	L
N	O	S	E	M	A	X	Z	R	U
B	R	E	N	F	G	Y	O	A	N
X	T	H	H	R	T	G	U	V	G
N	E	Q	A	A	B	E	F	K	S
I	N	H	A	L	E	N	V	Q	T
N	Y	R	E	X	H	A	L	E	B
W	I	N	D	P	I	P	E	L	H

Creativity 6





Food Making in Plants

All living beings need air, water and food to live. Food gives us energy to work. Plants also need energy. Plants make their own food. In this chapter, we take a look at how plants make food.



Plants Make Their Own Food

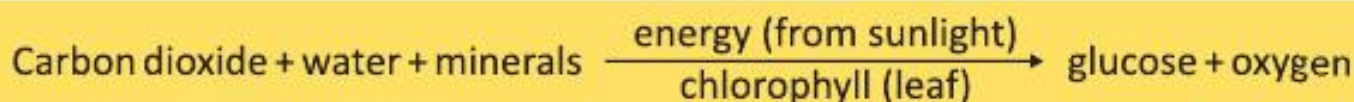
Why do plants have green leaves? They have green leaves because the leaves contain a substance called **chlorophyll**. This gives the leaves their green colour. Only green plants can make food because chlorophyll is needed to make food.



A plant gets **water** and **minerals** from the earth and **carbon dioxide** from the air. The energy needed to make food is taken from **sunlight**. The chlorophyll in the leaves traps sunlight. Leaves make food by the process called **photosynthesis**.

Photo means **light** and **synthesis** means **putting together**.

In the presence of sunlight, green leaves mix water, minerals and carbon dioxide together to make food. This plant food is a kind of sugar called **glucose**. When leaves make food, they give out **oxygen**. This process of making food is called **photosynthesis**. It is written as :



Green leaves are called food factories or kitchen of plants because they make food. We usually eat those parts of plants that have food stored in them.



flower



seeds



fruit



root



underground stem



leaves



Insectivorous Plants

How do non-green plants get food?

Some soils are poor in minerals, especially nitrogenous soil. Plants of these areas feed on insects to get the minerals they need. These plants are called **insectivorous plants**. Such plants are generally green in colour. So, they prepare their own food by using things like water, carbon dioxide and light.

Venus flytrap

The leaves of this plant are modified into two lobes that join at the centre. The edges have long, sharp spines. Together they form a trap on the inner surface of the leaf. There are hairs inside the trap. The plant secretes sweet fluid that attracts insects. When an insect lands on the leaf, it touches the hairs. The lobes close to trap the insect inside. The spines do not allow the insect to escape. Glands on the leaf surface then secrete digestive juices that help the plant to digest the insect. The trap then reopens.



Pitcher plant

The leaves of the pitcher plant are modified into a pitcher. The apex of the leaves forms a lid over the pitcher, that can be opened or closed. The pitcher has hairs inside it. When an insect sits on the pitcher of the plant, it closes its lid. The insect slips inside and gets trapped in the hairs. Inside the pitcher, digestive juices are released, that help the plant to digest the insect.



Moulds and **Mushrooms** are non-green plants they do not have chlorophyll. So, they cannot make their own food. They get their food from dead and decaying plants and animals.

BOTTOM LINE

Cuscuta or dodder plant commonly known as amarbel.

How do animals get their food?

Like plants, animals cannot make their own food. They obtain food from plants or animal sources. Based on the type of food they eat, animals are categorised into the following types—

- ▶ Animals that eat only plants or plant products, such as green leaves and grains are called **herbivores** or **herbivorous animals**. Animals such as cows, deer, goats and rabbits are herbivorous animals.
- ▶ Animals that eat flesh are called **carnivores** or **carnivorous animals**. Animals like lions, tigers, leopards, frogs and wolves are carnivorous animals.
- ▶ Animals that eat both plants and animals are known as **omnivores** or **omnivorous animals**. Animals like crows, dogs and sparrows are omnivorous animals.



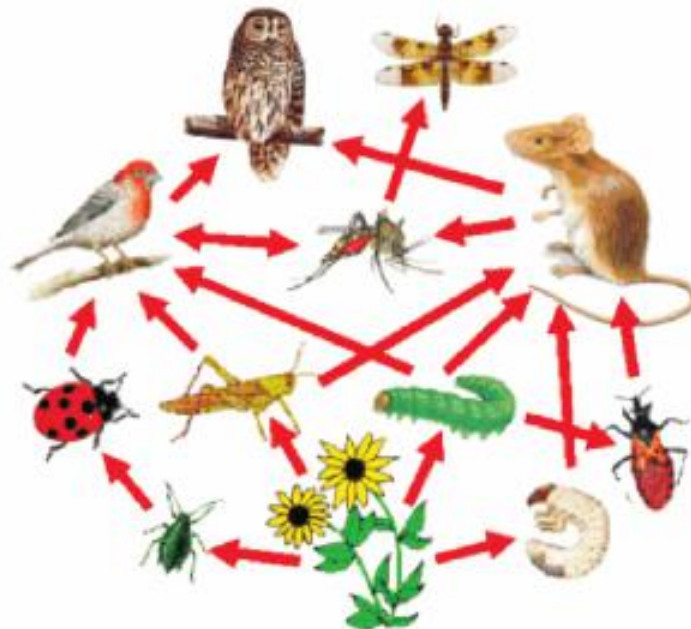


We have just learnt that food prepared by green plants is eaten by herbivores, who in turn are eaten by carnivores.

The process of transfer of energy in the form of food through a series of organisms is called a **food chain**.



Many animals eat more than one kind of food. So, different kinds of linked food chains exist in nature. Food chains do not operate in isolation. The network of inter-connected food chains is called a **food web**.



Interdependency between plants and animals

Plants and animals depend on each other for survival. Plants give food and oxygen to animals. Animals breathe out carbon-dioxide, that is used by plants to make their food. For this reason, plants and animals depend on each other and live together.



- ▶ Leaves make food by a process called photosynthesis.
- ▶ Green plants make their own food. Non-green plants get their food from dead and decaying plants and animals.
- ▶ Animals can be herbivorous, carnivorous, and omnivorous.
- ▶ The process of transfer of energy in the form of food through a series of organisms is called food chain.
- ▶ The network of inter-connected food chains is called a food web.



A. Multiple choice questions.

Tick (✓) the correct option :

1. _____ are called the Kitchen of the plant .

(a) leaves		(b) flowers	
(c) stems		(d) roots	
2. Animals that eat both plants and animals are known as _____.

(a) carnivores		(b) omnivores	
(c) herbivores		(d) mushrooms	
3. _____ is a non-green plant that gets its food from dead and decaying organisms.

(a) pitcher plant		(b) venus flytrap	
(c) mushroom		(d) none of these	
4. We eat the leaves of plants such as _____.

(a) cauliflower		(b) carrot	
(c) cabbage		(d) radish	

B. Fill in the blanks :

1. Extra glucose changes into _____ and is stored in plants.



2. Leaves of plants make food by a process called _____.
3. The network of inter-connected food chains is called a _____.
4. _____ and _____ depend on each other for survival.

C. Answer the following questions :

1. Why are leaves of plants green?

2. How does the pitcher plant catch insects?

3. What is a food web?

4. What is photosynthesis? Explain the process with diagram.

5. What is a food chain?

Extra Curriculum



■ **Circle the odd one in each group :**

- | | | | | |
|----|-------|------------|---------------|-------------|
| 1. | mould | mushroom | bacteria | sunflower |
| 2. | neem | rose plant | pitcher plant | lotus plant |

Subject Enrichment

■ **Draw a diagram of simple food chain.**

Art Integration





People at Work

People do different kinds of work to earn money. The work they do is called their occupation or profession. Let us meet few people at work to take a look at the jobs they do and the tools they use. By reading about them, we will also know how and where they learnt their skills.



Nitesh is a painter. He paints houses beautifully. He uses different brushes and colours to do his work. He learnt the art of painting from a building contractor.



Dr. Anurag Mishra is our family doctor. He treats people who are ill or injured. I also visit his clinic when I am sick. He uses a stethoscope. He keeps the stethoscope on my chest to listen to my heartbeat. He gives me medicine to cure my sickness. I recover soon after taking the medicine. He went to a medical college to learn how to do his job.





Naveen Sharma is an architect. He draws plans for buildings. A plan is a drawing of all the rooms in a building. The drawing shows where each room will be built in the building. He designed our house as well. He went to the college of architecture to learn to draw up plans for all sorts of buildings. He uses various tools to do his job.



John is our school bus driver. He drives our bus slowly and safely. He has a driving licence. He has learnt to drive the bus from his elder brother who is also an expert driver. He got his driving licence after passing a driving test.



Miss Anu is our class teacher. She teaches us social studies with a lot of love and patience. She is an excellent teacher. She uses various charts, globes and models to explain the lessons to us. She learnt to teach in a teaching college. Her mother is also a teacher.





Danish is a carpenter. He makes doors and windows. He also makes cupboards. He uses a **saw** to cut the wood. To make wood flat and smooth, he uses a plane. He uses a **drill** to make holes in the wood. He uses a **screwdriver** to put screws in the wood. He uses a **hammer** to fix nails in the wood. He learnt the art of carpentry from his uncle.



Mahesh Kumar is an electrician. He fits wires, switches, bulbs and fans. He repairs electrical appliances as well. The tools he uses are a drill, screwdriver and a pair of pliers. He learnt to do his work at an industrial training institute known as ITI.



Jeevan Das is a gardener. He looks after our school garden. He plants various flowering plants to make our school garden look beautiful. He uses a spade to dig the mud. He uses a trowel to remove unwanted plants. He uses a cutter to trim the plants. He learnt the art of gardening from his father.





Sanjay Mathur is a mechanic. He repairs scooters and motor cycles. He also mends punctures. The tools he uses includes a screwdriver, spanner, wrench and a pair of pliers. After finishing his schooling, he learnt to do his job at a motor repairing company.



REWIND

- ▶ Different kinds of works known as occupation are done by different people.
- ▶ Different occupations include mechanic, electrician, carpenter, doctor, driver, gardener, painter, architect etc.



A. Multiple choice questions.

Tick (✓) the correct option :

1. A person who makes plans for buildings is called an _____.
(a) air-hostess (b) engineer
(c) architect (d) doctor
2. To check our heartbeat, the doctor uses a _____.
(a) thermometer (b) stethoscope
(c) computer (d) saw
3. Charts and models are used by _____.
(a) carpenter (b) gardener
(c) teacher (d) driver



4. A gardener uses a spade to dig the _____.
- | | |
|-------------|------------|
| (a) plants | (b) mud |
| (c) flowers | (d) leaves |

B. Fill in the blanks :

1. A painter uses different kinds of _____ to paint.
2. A driver has to pass a driving test to get a _____.
3. _____ treats people who are ill or injured.
4. _____ repairs scooters and motorcycles.

C. Answer the following questions :

1. What is the work of an architect?

2. What is the job of an electrician?

3. Name the tools used by a carpenter and a painter.

4. Write the work done by these people :

(a) Doctor

(b) Tailor

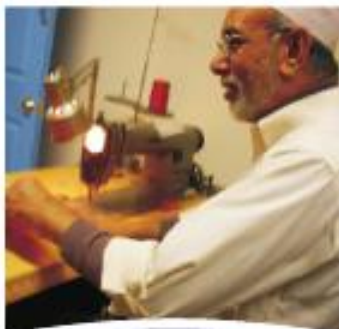
(c) Cobbler

(d) Gardener

Extra Curriculum



- Look at the pictures and write the names of the occupations in the boxes : **Subject Enrichment**











■ Take a look at the pictures given below and write their names.

Inter Disciplinary 6





Heritage of India

India is known for its monuments and ancient temples. These imposing buildings from the past are of great importance to us because they remind us of our glorious past and rich heritage. Some of the prominent monuments of our country are :



The **Qutub Minar** (minar means **tower**) is the highest stone tower of India. It is **Delhi's** most prominent landmark. It is made of red sandstone and is 72.5 metres high. Work on the construction of this tower was started by a king named **Qutub-ud-din Aibak**.

After his death, the minar was completed by **Iltutmish** who was the next king.

It is said that the Qutub Minar originally had seven storeys, but now only five remain. Each storey has a balcony. There is a narrow spiral staircase that goes to the top. The walls bear inscriptions from the **Holy Quran**. It has 379 curved steps. Near the Qutub Minar, there is an **Iron Pillar**. It has remained **rust-free** for the last 1,500 years!

It is also considered that it was built by Raja Vikramaditya in 5th Century by the name the Surya Stambha. It is an observatory to study astronomy.



Taj Mahal

The most famous monument of India is the **Taj Mahal**. It was built at Agra by the Mughal emperor **Shah Jahan** in memory of his beloved wife **Mumtaz Mahal**. It is made up of **white marble**. The walls have beautifully carved **floral designs**.

Once, it was decorated with semiprecious stones. It has huge garden with a reflecting pool. It took twenty thousand labourers and craftsmen and twenty years to build the Taj Mahal. It looks magnificent especially on moonlight.



In 1983, the Taj Mahal became a UNESCO World Heritage Site.

Every day thousands of tourists visit the Taj Mahal.

Char Minar

Situated at **Hyderabad**, the **Char Minar** is more than 400 years old. It was built by **Sultan Muhammad Quli Qutb Shah** to celebrate the eradication of the **plague** epidemic from the city.

This rectangular, four-arched structure is 100 feet high. It has four beautifully carved minarets at the four corners (each 53 metres high).



Meenakshi Temple



The **Meenakshi Temple** is the most famous temple of **Madurai**. This temple has two shrines, that are dedicated to **Meenakshi** (Goddess Parvati) and **Sundareswara** (Lord Shiva). In the Meenakshi Temple, there is a hall that has **one thousand pillars**. It is suggested to built about 2000 years ago.

Scenes from the lives of gods and goddesses are carved on these pillars. The brightly coloured **gopuram** or **tower** can be seen from a distance.



Hawa Mahal



Hawa Mahal is a beautiful palace. It is also known as the **Palace of winds**, mainly because the building has small balconies that maintain a constant flow of cool breeze. It was constructed so that the women of royal families could watch the processions and observe crowd passing through the streets being noticed by the people.

Situated at Jaipur, this palace was built by Maharaja **Sawai Pratap Singh**. It now houses a small museum.



India Gate

The **India Gate** of Delhi was built to pay homage to the soldiers who lost their lives during the first world war. The names of the soldiers who died fighting for the country are inscribed on the walls of the gate. There is a shrine under the arch of the gate. The **Amar Jawan Jyoti** has been burning without fail in this shrine since 1971 in the memory of these soldiers.



Sun Temple



The **Sun Temple** is situated at **Konark**, near **Puri** in **Orissa**. This grand temple is dedicated to the Sun God. The main temple is in the shape of a **chariot** with twelve pairs of huge **wheels** symbolizing twelve months driven by **seven horses**. Symbolizing seven colours of sun light which is white and seven days of the week.

Around 1200 craftsmen and labourers worked for 12 years to build this temple. It was built in 1250. It was declared by UNESCO a World Heritage Site.





Gol Gumbaz

The **Gol Gumbaz** at Bijapur, Karnataka is the second largest dome in the world. It is a memorial tomb built for a ruler, **Muhammed Adil Shah**. This huge dome stands without any pillar.

There is a gallery inside the dome. If you whisper, the gallery sends back faint echoes. That is why this gallery is called the **whispering gallery** of the **Gol Gumbaz**.



Monument's Preservation

The government of India has established the **Archaeological Survey of India (ASI)** to protect and preserve our historical monuments. The ASI also undertakes the **repair and maintenance** of the monuments.

Damaging these monuments is a punishable offence.

We can take care of our monuments in the following ways :

- ▶ Never scratch, write or draw on the walls of monuments.
- ▶ Not throw any garbage in the area around the monument.
- ▶ Always purchase the entry ticket when we visit a monument. This money is used for the maintenance of the monument.
- ▶ Not steal or destroy status or any other object of historical significance from these monuments.
- ▶ Do your best to reduce air pollution in the area surrounding the monument or at least make the people aware of it.

REWIND



- ▶ India is known for its monuments.
- ▶ Prominent monuments of our country include Qutub Minar, Gol Gumbaz, Hawa Mahal, Meenakshi Temple and so on.
- ▶ We should preserve our historical monuments.





A. Multiple choice questions.

Tick (✓) the correct option :

1. Shahjahan built the _____.
(a) Qutub Minar (b) Hawa Mahal
(c) Taj Mahal (d) Gol Gumbaz
2. Char minar is located in _____.
(a) Delhi (b) Hyderabad
(c) Konark (d) Kolkata
3. The _____ is the most famous temple of Madurai.
(a) Sun Temple (b) Meenakshi Temple
(c) Lotus Temple (d) none of these
4. India Gate was built to pay _____ to the martyrs of the first world war.
(a) homage (b) honour
(c) respect (d) none of these

B. Fill in the blanks :

1. The _____ is situated at Konark.
2. _____ is also known as the Palace of winds.
3. The _____ is the highest stone tower of India.
4. The full form of ASI is _____.

C. Answer the following questions :

1. Why are historical monuments important to us?

2. Why was the India Gate built?

3. Why did Shah Jahan build the Taj Mahal? Describe the beauty of this monument.



4. Why is the gallery of the Gol Gumbaz known as the whispering gallery?

5. Describe the structural beauty of :

(a) Hawa Mahal _____

(b) Sun Temple _____

(c) Qutub Minar _____

(d) Meenakshi Temple _____

Extra Curriculum



■ Write the names of the following pictures :



Subject Enrichment 

■ Make a plan and visit museum situated in New Delhi and write about your visit in five lines in your notebooks.

Experimental Learning 





National Symbols

Our country is India.

We love our country.

India is our motherland. We all are citizens of India.

We are proud of our country.



Our national symbols are the things through which we associate with our country. We should respect our national symbols.

National Flag

Our national flag is called **Tricolour**.

It has three stripes of different colours. The top stripe is **saffron** in colour. It stands for sacrifice. The middle stripe is white in colour. It stands for peace. The bottom stripe is green in colour. It stands for prosperity. There is a wheel in the centre. It is navy blue in colour. It has 24 spokes representing the 24 hours of the day.



We hoist our national flag on the Independence Day and the Republic Day.

We should respect our national flag.

National Emblem

Our national emblem has four lions. We can see three lions at a time. The fourth lion

is at the back. At the bottom there is a 'chakra'. This chakra can also be seen on our national flag. There is a 'horse' (represents loyalty, speed and energy) to the left side and a 'bull' (represents hard work) to the right side of the chakra.

National Flower

Lotus is our national flower. It's colour is pink. It is a very beautiful flower.

National Bird

Peacock is our national bird. It is a very beautiful bird. Lord Krishna likes it very much. Peacock lives in jungle and mostly seen in garden.

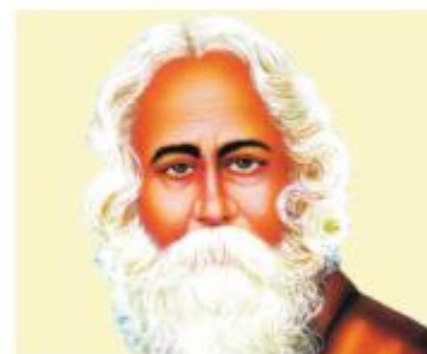
National Animal

Tiger is our national animal. It displays strength, agility and power.



National Anthem

Our national anthem is 'Jana-gana-mana.....'. It was written by **Rabindranath Tagore**. We should always stand when it is played. We should always keep silent at the singing time.



National Song

Our national song is 'Vande Mataram.....'. It was written by **Bankim Chandra Chatterjee**. We should love our national song.





- ▶ Our national flag is called Tricolour.
- ▶ Our national flower is lotus.
- ▶ Our national bird is peacock.
- ▶ Our national animal is tiger.
- ▶ We should respect our National Symbols.



A. Multiple choice questions.

Tick (✓) the correct option :

1. Our National Anthem was written by _____.
 (a) Bankim Chandra Chatterjee
 (b) Rabindranath Tagore
 (c) None of these
2. Our national flower is _____.
 (a) rose (b) jasmine (c) lotus
3. Our national bird is _____.
 (a) peacock (b) parrot (c) pigeon
4. Our national animal is _____.
 (a) lion (b) tiger (c) leopard

B. Fill in the blanks :

1. Our _____ is called Tricolour.
2. The wheel of Tiranga has 24 _____.
3. Lotus is our _____.
4. _____ is our national animal.

C. Write 'T' for true and 'F' for false :

1. We stand when the national anthem is played.



2. 'Tricolour' is our national emblem.
3. The wheel of Tricolour is red in colour.
4. We should respect our national flag.



D. Match the column A with column B :

A	B
1. Chakra	(i) tiger
2. Green colour	(ii) peacock
3. National flower	(iii) 24 spokes
4. National bird	(iv) prosperity
5. National animal	(v) lotus

E. Answer the following questions :

1. What is 'Tricolour'?

2. Who wrote our national anthem?

3. Who wrote our national song?

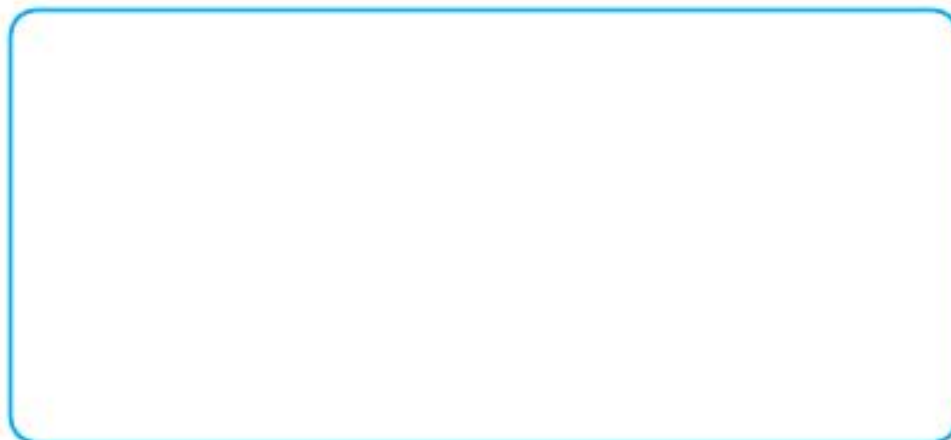
4. Explain the three colours present in the national flag.

Extra Curriculum



- Make a picture of our national flag and colour it.

Art Integration 

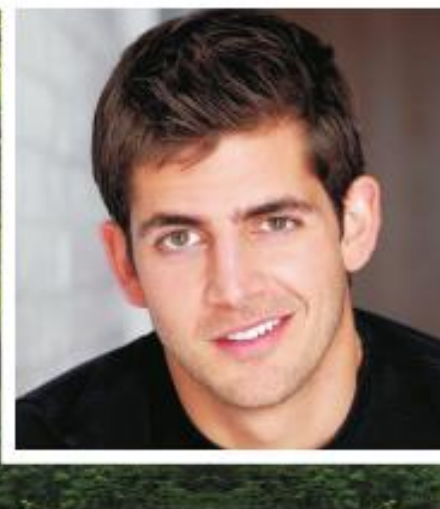




Forests

A forest is a large area of land thickly covered with trees and shrubs.

Forests provided the early man firewood, nuts and protection from weather. They also provided logs for rafts and materials for making shelter, utensils and weapons.



Forests serve modern man in many ways :

- ▶ Forests are greatly responsible for rainfall. Tree leaves transpire water. This leads to cloud formation and consequently rain.
- ▶ Forests provide us wood. Furniture, boats, wagons, sledges and countless other things are made of wood. An important use of wood is the production of paper.
- ▶ Forests are the 'lungs' of the environment. They clean the air for us.



- ▶ Forests are the homes to a variety of wildlife.
- ▶ Forest animals provide us meat, honey, lac, skin and many other useful things.
- ▶ Forests provide us many products of economic importance such as bamboos, canes, resins and essential oils.
- ▶ Forests prevent soil-erosion. The roots of trees bind the soil and do not allow it to be washed away by rainwater or blown away by the wind.



Forest People

India is a big country. Different states have different tribes. They differ from one another in their cultures, languages, handicrafts, customs and religions.

Forests are homes to many tribal people.



Products from Forests

Medicinal plants

Medicinal plants like cinchona, neem and amla are used for curing diseases.

Fodder

Grass is used as fodder for domestic animals.

Wood

Wood is also used as a fuel for cooking and for heating homes during winter.

Building materials

Wood is used to construct houses. Leaves and grass are used to make roofs of these houses.



Food

We get fruits like **custard apple**, **tamarind**, **wood apple** (bel), **amla** and **edible berries** as food.



Forests also provide us with a variety of **nuts**.

Spices

Spices like cloves and cinnamon are used in cooking.

Forest products

Tribal people collect :

- ▶ **Bamboo** to make baskets.
- ▶ Grass to make brooms and mats.
- ▶ Rubber juice from **rubber** trees.
- ▶ Leaves to make pattal.



Sacred Groves

Many tribal societies in our country worship nature.

They worship many plants and animals. The **peepal**, **banyan** and **neem** trees are considered sacred trees. At religious festivals, these trees are decorated and worshipped. It is believed that such acts bring prosperity, health, good harvest and peace.

Sacred groves are an important part of some tribal societies. A patch of forest land is dedicated to a deity or village god. This piece of land is considered as **sacred grove**. No one can cut a tree or kill an animal or a bird in this sacred area. Local tribal people protect sacred groves and in doing so, promote conservation of forests.





Depletion Of Forests

Cutting down of forests is called **deforestation**. Deforestation causes soil erosion and makes the land infertile. Landslides and floods are common in deforested areas. Climate also becomes warmer. Animals lose their homes when we cut down forests.



Causes Of Deforestation

Urbanisation

The growth of towns and cities causes deforestation in many parts of the world. The growing demand for fire wood in villages and timber in urban areas is also a major cause of deforestation.

Shifting cultivation

In many forests, tribal people clear a patch of forest land by cutting trees. They use this land to cultivate crops. After a few years, due to not using manures or fertilizers, the land becomes barren and unfit for cultivation. They then leave this land and clear another patch of forest land for cultivation.



Deforestation

Excessive cutting down of trees has robbed tribal people of their means of livelihood. Many of them have been forced to leave their habitat and settle in nearby villages and towns. These people find it hard to adjust themselves to the new surroundings.

They want to work, but they cannot find employment, because they are uneducated and unskilled.



Forest Conservation

We need to protect our forests. Now-a-days forests cannot be cleared without the permission of the government.

▶ A large number of trees are planted with the help of the local people to bring wasteland under plantations.



- ▶ Cutting down of trees is totally banned for lumbering companies.
- ▶ Tribal people cut trees to earn their livelihood. But they plant trees and look after them till they are fully grown.

So, the balance in nature is not disturbed.

REWIND



- ▶ We should take care of our trees and forests.
- ▶ Forests are our wealth.
- ▶ We should protect our forests.
- ▶ People should be made aware of the importance of forest conservation.



A. Multiple choice questions.

Tick (✓) the correct option :

1. By cleaning the air for us, forests are called the _____ of the environment.

(a) heart	(b) kidneys
(c) lungs	(d) eyes
2. We get rubber from juice the _____ tree.

(a) eucalyptus	(b) cinchona
(c) neem	(d) rubber
3. Growing crops by clearing forest land is called _____.

(a) deforestation	(b) afforestation
(c) shifting cultivation	(d) none of these

B. Fill in the blanks :

1. The _____ of trees help to prevent soil erosion.
2. Tribal people make _____ from bamboo.



3. A patch of forest land dedicated to deity is _____.
4. _____ is used to make baskets.

C. Answer the following questions :

1. Why are forests called the lungs of the environment?

2. How are sacred groves protected by tribal people?

3. How do forests help to prevent soil erosion?

4. What are the reasons for deforestation?

5. List three ways in which tribal people depend on forests.

Extra Curriculum



- Paste at least three pictures of the things we get from forests and name them.

Subject Enrichment 





Wildlife



Wildlife refers to any wild animal or plant living in its natural habitat. There are more than 5,000 species of animals and 1,200 species of birds found in Indian forests.

Our rich and varied wildlife is our greatest national heritage.

The famous Bengal tiger has its home in the Sundarbans of West Bengal.



Monkeys and elephants are found in the forests of Kerala, Karnataka and Tamil Nadu.

When you visit **Kaziranga National Park** (Assam), you see the **one-horned rhinoceros**.

The **Gir National Park** in Gujarat is home to majestic lions.





Decreasing Numbers Of Wild Animals

Excessive hunting

Early humans lived in the forest. They killed wild animals for food. Today, humans kill animals not just for food.

For example : Animal like leopards, tigers, crocodiles and snakes are being killed for their skin. Their skin is used for making leather goods. The Indian rhino is being killed for its horn as it is said that it has some medicinal properties.



Deforestation

Due to an increase in population, more and more trees are being cut down to make room for industries, agricultural farms and cities. Cutting down trees has led to the destruction of the habitat of many wild animals.



Conservation Of Wildlife

Animals are our national heritage and must be protected. Tough measures to save wildlife have been introduced.

For example : Hunting of animals is completely banned. It is a punishable offence.

A large number of **national parks** and **wildlife sanctuaries** have been set up in different parts of the country.

These places are reserved and protected areas where animals and birds live freely.

Preventing deforestation is another way of preserving wildlife.

Media, such as radio and television, are being used to **create awareness** among people about the importance of wildlife.





Project Tiger

When the famous **Bengal tiger** of **Sundarbans** was threatened with extinction, the government started '**Project Tiger**'. Under this programme, tigers in various parts of India were given protection. This programme proved to be very effective. Currently, 27 tiger projects are being run in 14 states which has resulted in the increases of number of tigers.

There is also an another project i.e. the Rhino project. Rhinos are being saved from extinction at the **Kaziranga National Park**.

We help to conserve wildlife by refusing to buy things made of ivory, horns and animal skin.

BOTTOM LINE

Wildlife week (October 1 to October 8) is observed every year with a view to educate people on the importance of wildlife.

REWIND



- ▶ Wildlife refers to any wild animal or plant living in its natural habitat.
- ▶ We should protect our wildlife so that we can pass our national heritage on to future generations.
- ▶ Deforestation and excessive huntings are the example of decreasing numbers of wild animals.



A. Multiple choice questions.

Tick (✓) the correct option :

- Elephants are found in the forests of _____.
(a) Gujarat (b) Kerala
(c) Kashmir (d) Karnataka
- At Kaziranga National Park, we see _____.
(a) one-horned rhinoceros (b) lions
(c) tigers (d) deer



3. The Gir National Park in Gujarat is home to _____ .
 (a) tigers (b) lions
 (c) rhinoceros (d) leopards
4. _____ is a reserved area where birds and animals are protected in their natural surroundings.
 (a) forest (b) national park
 (c) zoo (d) none of these

B. Fill in the blanks :

- Rhinoceros is being killed for its _____ .
- Animals are our national _____ and must be protected.
- _____ refers to any wild animal or plant living in its natural heritage.
- Our rich and varied wildlife is our greatest _____ .

C. Answer the following questions :

- How is deforestation a threat to wildlife?

- Name the national park for rhinos.

- What is the Project Tiger?

- What are the steps taken by the government to conserve wildlife?

- Why are national parks set up in our country?

Extra Curriculum



■ **Circle the odd one :**

- | | | | |
|--------------|----------|------------|-------------|
| 1. crocodile | snake | rhinoceros | leopard |
| 2. tiger | lion | eucalyptus | rhinoceros |
| 3. bison | antelope | tiger | sikkim stag |

- Collect at least three pictures of different wild animals and paste them in your scrapbook.

Computational Thinking

Creativity





Pollution

Pollution is the addition of harmful substances in air, water and land. These harmful substances are known as **pollutants**. There are four kinds of pollution : **air**, **water**, **noise** and **soil**.



Water Pollution

Water pollution is caused due to both natural and man-made factors. An important natural polluting factor is storm. Storms usually cause mud, soil and other harmful particles to dissolve in water. In this manner, water becomes unfit for consumption.

Water pollution is also caused by human activities.

The most common and widely prevalent human activities that cause water pollution are :

- ▶ Leakage from oil tankers.
- ▶ Chemical waste from factories, household waste and sewage being dumped into ponds, rivers and lakes.
- ▶ Washing clothes and bathing cattle in rivers, streams, ponds and lakes.

Drinking polluted water causes diseases like jaundice, typhoid and gastroenteritis. Water pollution affects not only humans but plants and animals too. Plants and animals that live in or use polluted water become weak and slowly die. Due to the presence of pollutants in water, plants are unable to prepare their food. Their growth got affected. Animals consume this polluted water and become sick.





Noise Pollution

Loud noise is a pollutant that cause noise pollution. **Noise pollution** is also known as **sound pollution**. Noise pollution is harmful for all living organisms.

Noise pollution is caused by noise from :

- ▶ Cars, aeroplanes and buses
- ▶ Loud music
- ▶ Loudspeakers
- ▶ Machines in factories

Noise pollution causes irreparable damage, it causes.

- ▶ Stress, irritation, headache
- ▶ Loss of hearing and in extreme cases deafness
- ▶ Loss of sleep



Soil Pollution



Soil pollution is caused when solid wastes are dumped on the land.

Soil pollution makes the soil infertile and is harmful to all living things. It also affects the growth of plants.

The main causes of soil pollution are :

- ▶ Continuous use of pesticides and fertilizers on a piece of land.
- ▶ Industrial and household waste being dumped on the land.



Air Pollution

Nature and humans cause **air pollution**. Nature pollutes the air by sandstorms, eruption of volcanoes, forest fires and pollen from flowers.

Most air pollution is caused by human activities. Burning of fuels in factories and vehicles, burning of garbage, release of smoke and harmful gases into the atmosphere are a few activities that cause air pollution.



Air pollution causes increase in a number of diseases, breathing problems and lung damages. Plants and animals are also affected by air pollution. Photosynthesis in plants got affected by polluted air. Plants are not able to prepare their food properly. This affects their growth. In extreme cases, plants die.

Thus, we understand the pollution is a danger looming large over the earth.

We must remember that almost all pollutions are caused by human activities either directly or indirectly— use of mosquito repellents, pesticides, air conditioners and loud speakers are just a few examples of pollutants.



REWIND



- ▶ Pollution is the addition of harmful substances in air, water and land.
- ▶ There are four main kinds of pollutions : water pollution, noise pollution, soil pollution and air pollution.
- ▶ Pollution is a danger looming large over the earth.
- ▶ Small acts such as Reusing, Recycling, being careful about small acts such as avoid making loud noise, planting a sapling or throwing garbage into the garbage bin will certainly help to save our home i.e. our earth.



A. Multiple choice questions.

Tick (✓) the correct option :

1. There are _____ kinds of pollution.
- | | | |
|----------|--------------------------|-----------|
| (a) one | ☐ | (b) three |
| (c) four | ☐ | (d) five |



2. Harmful substances in the environment are called _____.

(a) pollutants

(b) wastes

(c) garbage

(d) pollution

3. Aeroplanes cause _____ pollution.

(a) water

(b) soil

(c) noise

(d) air

4. High use of fertilizers causes _____ pollution.

(a) noise

(b) soil

(c) water

(d) air

B. Fill in the blanks :

1. Drinking polluted water causes diseases like _____ and _____.

2. Noise pollution causes _____ and _____.

3. Nature pollutes the air by _____ and _____.

4. _____ pollution is caused when solid wastes are dumped on the land.

C. List two effects each of :

(a) Noise pollution

(b) Air pollution

D. Answer the following questions :

1. What does pollution mean?

2. How does water get polluted?

3. List two causes of soil pollution.

4. How can you reduce pollution?





- Write four sentences to describe 'Pollution and ways to control it.'

1. _____
2. _____
3. _____
4. _____

- Name the types of pollutions and paste pictures of each of them in the space provided.





Fuels



We eat food to get energy to work. Vehicles such as cars, motorcycles, ships, trains, aeroplanes etc. also need energy to move.



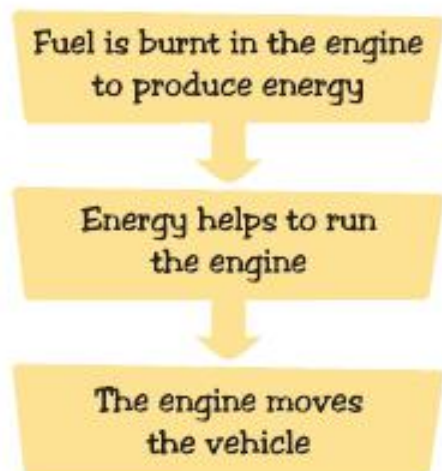
The materials that these vehicles burn to produce energy are called **fuels**. Modern day vehicles use different types of fuels.



Today, almost all forms of transport such as cars, buses, scooters, aeroplanes and ships depend on petroleum products like petrol, diesel and CNG (compressed natural gas).

All motor vehicles have an engine.

This is how an engine runs.

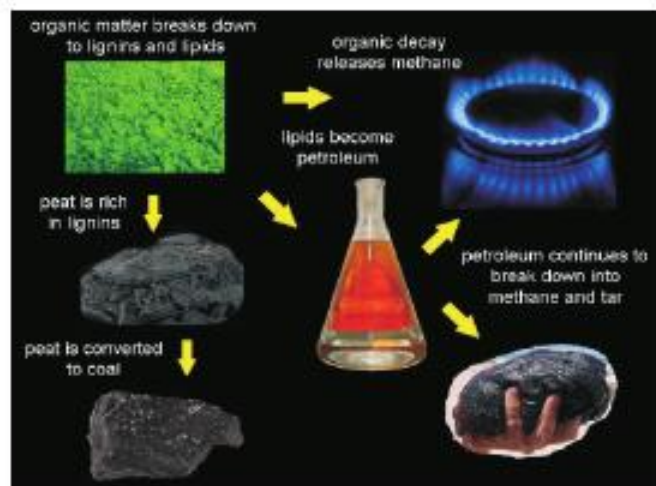


Formation of petroleum and its products

Petroleum is a dark-coloured thick crude oil. In Latin, '**Petra**' means 'rock' and '**oleum**' means 'oil'.

So, the word '**Petroleum**' means 'oil from rocks'.

- ▶ Petroleum oil was formed when small plants and animals got buried under the sea millions of years ago. Due to high pressure and temperature, their bodies were slowly converted into petroleum.
- ▶ Petroleum is obtained by drilling holes called **oil wells** into the earth's crust.
- ▶ The petroleum so obtained is taken to an oil refinery where, products such as petrol, kerosene, diesel, fuel oil and fuel gas are obtained from crude petroleum.



How do we get petrol and diesel?

- ▶ The petrol and diesel obtained at the oil refinery are supplied to the petrol depots in different parts of the country by trains and trucks.
- ▶ From these depots, petrol is supplied to the petrol pumps by means of trucks.
- ▶ We get petrol and diesel from these petrol pumps.

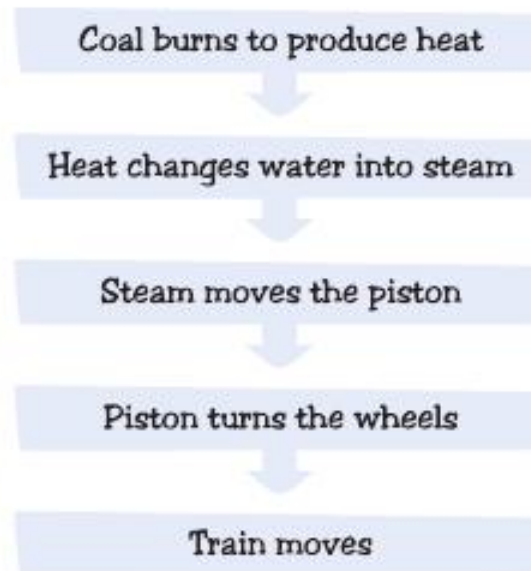




In the past, coal was used as a fuel to run steam engines and ships. In a steam engine, coal was burnt to produce heat. This heat boiled the water in the boiler to produce steam. Steam moved the piston, which in turn, turned the wheels of the engine.



This is how a train moves :

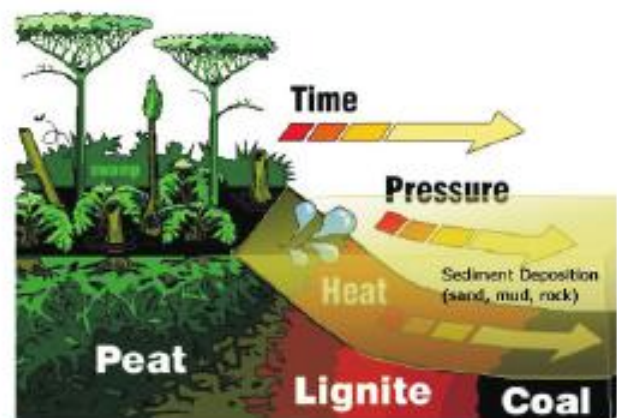


Coal is a valuable natural resource. It is a sedimentary rock formed from the remains of plants.

Millions of years ago the earth was covered with thick forests growing in swamps

Due to the earthquakes and volcanic eruptions many forests were buried under the surface

Due to high temperature and pressure inside the earth, the remains of the plants turned into coal





Uses Of Coal

- ▶ Coal is used as a fuel mainly for cooking, producing electricity and moving engines.
- ▶ Many chemicals made from coal are used in drugs, plastic, nylon and fertilizers.



Conservation Of Fuel

We can conserve coal and petroleum in the following ways :

- ▶ We should use public transport like buses and metro trains to reduce the number of vehicles on the road. In this way, we can save a lot of fuel.
- ▶ We should avoid driving very short distances. Instead, we should **walk or use a bicycle**. This saves fuel.
- ▶ We should keep our vehicles in good condition. We should keep the engines of our vehicles properly tuned. Periodic servicing improves efficiency and reduces fuel consumption.



Non-renewable Sources of Energy

Fuels like coal and petroleum take millions of years to form. The rate at which they are being used indicates they are likely to get exhausted very soon. The sources of energy that gets exhausted quickly and cannot be renewed are called **non-renewable sources of energy**. **Example** : coal and petroleum.

Alternative Sources of Energy

The number of vehicles in our country are increasing day by day. High prices, limited supply and polluting effects of fuels like petrol and diesel have compelled us to search



for renewable sources of energy. These are also called **alternative sources of energy**. Alternative sources of energy such as **wind energy, solar energy and energy of flowing water** are renewable sources and do not exhaust. In addition, their use do not cause pollution.



REWIND

- ▶ The materials that the vehicles burn to produce energy are called fuels.
- ▶ Modern day vehicles use different types of fuels such as coal and petroleum.
- ▶ We can conserve coal and petroleum by periodic services, using public transport, walking or using a bicycle.



A. Multiple choice questions.

Tick (✓) the correct option :

1. Coal is a _____ rock.

(a) metamorphic	(b) sedimentary
(c) igneous	(d) all of these
2. We get petrol from a _____.

(a) petrol pump	(b) tube well
(c) river	(d) ocean
3. A renewable source of energy is _____.

(a) petrol	(b) coal
(c) sun	(d) moon



B. Fill in the blanks :

1. Regular servicing of our vehicles increase their _____.
2. The full form of CNG is _____.
3. The materials that are burnt to produce energy are called _____.
4. The sources of energy that are exhaustible are _____ and _____.

C. Write 'T' for true and 'F' for false :

1. Coal is a valuable natural resource.
2. We should not avoid driving very short distances.
3. All vehicles need fuel to run.
4. Wind energy is a renewable source of energy.



D. Answer the following questions :

1. What are fuels? Name any two fuels.

2. Name any two petroleum products.

3. What are non-renewable sources of energy?

4. How is coal produced in nature?

5. How can we conserve coal and petroleum?

6. What are the alternative sources of energy? Give three examples.

7. How was petroleum formed millions of years ago? How is it obtained from the earth's crust?





- Paste a picture of CNG vehicle in the space provided and also write five lines on it. Creativity 6

1. _____
2. _____
3. _____
4. _____
5. _____





Natural Disasters

A calamity is an event that brings terrible loss. It is a disaster. Natural calamities are calamities caused by nature. They include disasters such as earthquakes, floods, landslides, cyclones and volcanic eruptions.

Some calamities are predictable for example, a **hurricane**. Some calamities such as earthquakes, take us by surprise. It is always good for us to know about the different kinds of calamities so that we can prepare ourselves for these. We should also remember that during a calamity we should come forward and help the victims of the calamity.



A sudden, violent shaking or trembling of the earth's surface is called an **earthquake**.

Earthquakes are caused by the collision of the plates of rocks under the surface of the earth. Earthquakes of great intensity cause a lot of damage to life and property.





Drought is a long period of time when there is little or no rain. It results in severe water shortage. Ponds and lakes dry up. Huge cracks develop in the soil. Due to scarcity of water, crops get damaged. People do not get food to eat. The animals starve as they depend mostly on the farm waste.



Due to heavy rains, rivers overflow and spread out onto the land around them. This event in nature is called **flood**. During floods, large areas of dryland gets submerged in water.



A fast-moving, violent and destructive wind-storm is called a **cyclone**. A cyclone can destroy houses and crops. A cyclone at sea brings huge volumes of water from the sea to coastal areas. The rushing sea water destroys everything that comes in its path. Cyclones are a great threat to people living in coastal areas.





When a massive rock breaks off and slides down a hillside or mountain side, it is called a **landslide**. It destroys everything that comes in its path.



A volcano is a hill with a vent. The vent reaches down to the hot, molten rock beneath the surface of the earth. During a **volcanic eruption**, the molten rock is forced up the vent. It flows **out of the top** as lava. The lava flows out very fast and burns everything that comes in its way. It causes great damage to the property and life.



Role of various communities during natural calamities

We cannot control natural forces and prevent natural calamities. But we can help the victims and try to reduce their sufferings. It is our moral duty to help one another in the times of trouble. Government agencies, communities and voluntary organisations do a lot towards bringing relief to people affected by natural calamities.

- ▶ Money, food items, clothes, blankets and medicines are donated by people and sent to the affected areas.
- ▶ Various organisations set up relief camps. They collect donations in many forms and help the affected people living in these camps.



Red cross society provides the affected people with all types of medical help. Many doctors and nurses look after the sick, hurt and the affected people.

- ▶ These organisations work hard to maintain hygienic conditions in the affected areas and in the relief camps to prevent the outbreak of an **epidemic**.
- ▶ They also organise shows and exhibitions to collect funds. Famous film stars and cricket stars participate in these shows to draw support. They also create an awareness among people about the suffering of the victims.



- ▶ Radio, television and newspaper plays an important role in helping the affected people. In their special reports and bulletins, they convey the messages of the affected people to their relatives living elsewhere.
- ▶ The telephone department provides emergency phone numbers and help line services at relief camps during such times of crises.

The other agencies that help people affected by natural calamities are :

- ▶ The Air force
- ▶ The Police
- ▶ The Navy

The **Air force** uses its helicopter to rescue people and drop food, water, medicines and other necessary items in the affected areas.



BOTTOM LINE

Cutting of forests leads to flooding. Due to the lack of trees, a greater quantity of rainfall is absorbed into the soil. This makes the soil wet and loose. Thus the soil gets washed off easily, leading to floods.

The **Police** works to save the people, protect their property and help them to reach relief camps safely.

The **Navy** bring in its boats.

We can protect ourselves from disasters if we start taking care of our environment.





- ▶ A calamity is an event that brings terrible loss.
- ▶ Examples of natural calamities are earthquakes, floods, landslides, cyclones and volcanic eruptions.
- ▶ Various communities like Red cross society, government agencies etc. help the people during these natural calamities.



A. Multiple choice questions.

Tick (✓) the correct option :

1. Shaking or trembling of the earth's surface is called an _____.
 (a) cyclone ☐ (b) landslide ☐
 (c) earthquake ☐ (d) drought ☐
2. Heavy rains can lead to _____.
 (a) volcanic eruption ☐ (b) floods ☐
 (c) earthquake ☐ (d) cyclone ☐
3. Huge cracks develop in the soil during _____.
 (a) drought ☐ (b) cyclones ☐
 (c) landslides ☐ (d) earthquakes ☐
4. _____ department provides emergency phone numbers at relief camps.
 (a) police ☐ (b) red cross ☐
 (c) telephone department ☐ (d) none of these ☐

B. Fill in the blanks :

1. A fast-moving and destructive wind-storm is called a _____.
2. During a volcanic eruption, the hot molten fluid flows out as _____.
3. _____ is a long period of time when there is little or no rain.
4. _____ provides the affected people with all types of medical help.



C. Answer the following questions :

1. What are natural calamities? Name any three.

2. What happens during floods?

3. Why do the people living near seashores afraid of cyclones?

4. How is an earthquake caused?

5. What happens during a drought?

6. What is the role of the community during natural calamities?

Extra Curriculum



■ **Circle the odd one.**

1. earthquake

flood

train accident

cyclone

2. radio

television

newspaper

textbook

3. navy

helicopter

airforce

police

Computational Thinking

■ **Name these natural calamities :**









Subject Enrichment

■ **Write at least five lines on tsunami.**

1. _____

2. _____

3. _____

4. _____

5. _____

Inter Disciplinary





First Aid and Safety

Accidents can happen anytime, anywhere without any warning. They usually happen when someone is careless. To avoid accidents we must follow safety rules.



First Aid



Whenever an accident happens, the injured person must be given medical aid before the doctors arrival or before the victim is hospitalized.

This immediate help given to an injured person is called **first aid**.

It is very essential and can save the life of the injured person.

The main aims of first aid are—

- ▶ to give some relief from pain as in the case of a fracture.
- ▶ to save a life, especially in case of poisoning.
- ▶ to prevent the injury from getting worse, especially if there is bleeding.



Kinds Of Injuries

Cuts and Wounds (Minor)

- ▶ Wash your hands thoroughly before giving first aid to the injured person.
- ▶ Remove all dirt from the wound and clean the





skin around it with small pads of cotton soaked in an antiseptic solution.

- ▶ Cover the wound with a clean cotton dressing to stop bleeding. Tie a bandage. Make sure that the bandage is not very tight.

Cuts and Wounds (Deep)

A deep cut bleeds heavily, it is important to stop the bleeding.

- ▶ First wash the wound with soap and plenty of water.
- ▶ Take some cotton or make a thick pad of sterile gauze and press it over the wound. Then tie it tightly with a bandage. If the wound does not stop bleeding, use a bandage called **tourniquet**.

It is a wide strip of cloth band which is tightly wrapped around the wound two or three times and then knotted.



- ▶ Raise the bleeding part with the tourniquet, higher than the rest of the body.
- ▶ A deep wound must be shown to a doctor.

The doctor has to stitch the wound if it is very deep, to close it. When the skin is cut, germs enter the body through the injury and cause a dangerous disease called **tetanus**.

The doctor gives an anti-tetanus injection to the injured person to protect him/her from tetanus.

Nose Bleeding

During summer, bleeding from the nose is a very common problem with children.

- ▶ Ask the patient to sit on a chair comfortably with his/her head seated back and arms folded above the head.



- ▶ Provide the patient an ice-pack or wet cloth to keep on his nose. An ice-pack is made by tying ice-cubes in a clean handkerchief.
- ▶ Ask the patient to breathe through his/her mouth.
- ▶ Also do not allow him/her to blow his/her nose.
- ▶ For heavy bleeding, consult the doctor immediately.



Sprain

Sometimes joints such as the ankle or the wrist get twisted and get damaged. The joint swells up and is very painful.

- ▶ Use a cold ice-pack.
- ▶ Rub an ointment lightly that relieves pain, on the sprained joint.
- ▶ A crepe bandage should be wrapped around the sprained joint to support the joint.
- ▶ A sprained joint should not be moved so that it heals faster.

Fracture

When a person falls down and gets hurt, his/her bones may get cracked. This is called a **fracture**.



It is a fracture when—

- ▶ The part of the victim's body that is hurt is in great pain and victim is unable to move the broken part.
- ▶ There is swelling around the area.

You cannot treat a fracture on your own. The only first aid you can give is to ensure that the fractured part is not moved anymore.

Follow these steps for first aid in case of a fracture—

- ▶ Keep the patient calm and comfortable and call the doctor immediately.
- ▶ Take a magazine, piece of cardboard or wood, or a pillow and tie it around the fractured part. This act as a **splint** and prevents movement of the broken bone.



- ▶ When a bone of the hand is broken, make a **sling** using a cloth or a bandage and support the arm on it. The sling gives support and prevents movement.
- ▶ When a bone in the leg is broken, tie a splint around the fractured leg. Do not allow the patient to move his/her leg at all.
- ▶ When the patient has to be moved, he/she should be carried on a stretcher.



- ▶ Get an X-ray of the fractured part immediately.

Burns

One gets burnt with hot objects such as fire, boiling water or milk, crackers and chemicals. Burns are very painful.

They are of two types

Minor

Severe

Minor burns

In case of minor burns the skin turns red and there is a burning sensation on the affected area.

You must give the following first aid—

- ▶ Keep the burnt area under cold running water for a few minutes. Then apply ice-cubes on the burnt area. Cooling the burn gives relief from pain.



- ▶ Apply an antiseptic cream, on the burnt part. This gives relief from the burning sensation.

Severe Burns

In case of severe burns; take the patient to the doctor immediately. Severe burns need special care. Sometimes blisters are found in the burnt part. They should not be pricked. An open blister becomes an open wound through which germs enter the body and



cause infection. Cover the burnt area with a clean piece of cloth to protect it from germs and also from dust particles and insects.



Chemical Burns

When the burn is caused by a chemical, wash the affected area with plenty of water to remove the chemical. Then give first aid as for the other burns.

Fire Accidents

A fire is often very destructive. It causes loss of life and property. Fires spread very fast. Hence it is important to control it quickly. We control all kinds of fires by—

- ▶ reducing the temperature of the burning substances.
- ▶ cutting off the air supply.

Fire Fighting

When paper or wood is on fire, it is extinguished by throwing water over it. This reduces the temperature of the burning substances.



When the fire has been caused by petrol it is extinguished by throwing sand or mud over it. This cuts off the air supply to the fire. Never throw water over burning petrol or kerosene.

If the person's clothes catch fire, put water on him. Ask him to roll on the ground. Cover him

with a blanket or an overcoat. This cuts off the air supply and the fire gets extinguished.





When the fire is caused due to an electric short-circuit, it can be extinguished by throwing sand or mud over it. Switch off the main supply immediately to prevent spread of fire. Throwing water over such a fire can be dangerous.

When this flowing water touches someone, he/she can get an electric shock.

Fire can also be extinguished by some special devices called fire **extinguishers**, we must learn to use them. They should be used instead of water and sand for extinguishing fire.



In case of big fire, we should shout for help immediately and **dial 101** on our telephone to call the **fire brigade**. It is important to remain calm over the telephone and give the correct address and important landmark of the place of fire.

REWIND

- ▶ The immediate help given to an injured person is called **first aid**.
- ▶ Kinds of injuries include : Minor cuts and wounds, deep cuts and wounds and nose bleeding.
- ▶ To avoid accidents we must be more careful. We must learn and practice first aid to be used during an emergency.





A. Multiple choice questions.

Tick (✓) the correct option :

1. When a person gets a sprain, use an _____.
(a) antiseptic ☐ (b) injection ☐
(c) ice-pack ☐ (d) all of these ☐
2. When a bone gets broken or cracked, it is called a _____.
(a) bleeding ☐ (b) fracture ☐
(c) sprain ☐ (d) nose bleeding ☐
3. A chemical burn, should be washed with _____.
(a) water ☐ (b) soap ☐
(c) antiseptic solution ☐ (d) shampoo ☐
4. When the fire has been caused by petrol, it is extinguished by throwing _____ over it.
(a) sand ☐ (b) water ☐
(c) petrol ☐ (d) none of these ☐

B. Fill in the blanks :

1. Immediate help given to an injured person is called _____.
2. Accidents usually happen when somebody is _____.
3. The doctor gives an _____ injection to the injured person with deep wounds.
4. Dial 101 to call the _____.

C. Answer the following questions :

1. What is first aid?

2. What are the two ways by which we control fires?

3. What should you do as first aid if someone gets a chemical burn?



4. Why should water not be used to extinguish a electric fire?

Extra Curriculum



- Circle the odd one in each group. Give the reason for your answer.

Computational Thinking

1. fracture sprain water burn

Reason _____

2. cotton tourniquet dettol fracture

Reason _____

- What kind of first aid do you give if a person—

Life Skills & Values

(i) has nose bleeding _____

(ii) has a fracture _____

(iii) gets a deep cut or wound _____

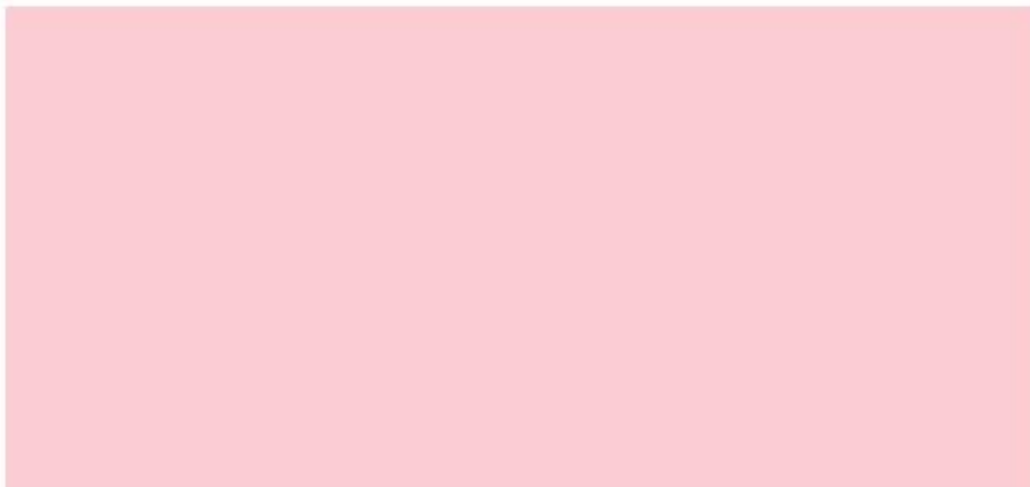
- Prepare a first aid box.

Creativity

You need a box to keep the things, some sterile cotton, doctor's tape, bandage, a pair of scissors, band-aids, antiseptic cream and liquid, cream for burns and sprains, a pain reliever, iodine and baking soda.

- Make the sign used for indicating Medical term.

Art Integration





Garbage and Its Disposal

Waste or garbage consists of materials that are no longer needed and are thrown away. A large quantity of waste is produced in houses, offices and factories. When all the waste is left in the open, it gives rise to diseases like dysentery, diarrhoea, typhoid and cholera. It also creates a foul smell. We must get rid of these wastes to keep our surrounding clean.



Types Of Wastes

Bio-degradable waste

Non-biodegradable waste



Bio-degradable Waste

Bury some vegetable peels and a coke can in the soil in a flower pot. Dig these things out after a week. You see that there is no sign of the **peels**. The can, however, is still there. Vegetable peels rotted away completely and became part of the soil. So we see, some types of waste decompose easily and become part of the soil.

This type of waste is called **bio-degradable waste**.

For example : Paper, leaves, waste food, grass and flowers are bio-degradable wastes.





Non-Biodegradable



Some of the wastes do not decompose easily. They do not become part of the soil. These wastes are called **non-biodegradable wastes**.

For example : Plastics, metals and glasses.

BOTTOM LINE

Non-biodegradable wastes stay unchanged in the soil for a long time and pollute the soil. When animals like cows eat these waste, they fall sick and at times even die.



Disposal Of Solid Waste

Wastes are disposed in three different ways :

Sanitary land-filling

Open dumping

Incineration



Sanitary Land-filling

In this method, big, deep pits called **landfills** are dug outside cities. The waste picked up by municipality trucks dump the waste in these landfills. These pits are covered with a thick layer of soil or a stone slab. This method is a better way of disposing off solid waste.

- ▶ The waste decomposes in the landfills and becomes a part of the soil.
- ▶ It does not provide any breeding ground for insects or germs. So there is hardly any risk of catching a disease.
- ▶ It does not create any foul smell.





Open Dumping



In this method, the waste is thrown openly to decompose with the action of air, water and sunlight.

This method is harmful because the dumping area becomes a breeding ground for insects and germs. This increases the risk of catching disease. In such unclean surroundings, diseases spread very fast. This method also creates a foul smell in the surroundings.



Incineration

In cities, solid wastes are sometimes burnt in big furnaces called **incinerators**. This method is called **incineration**. It does not allow insects or germs to breed. It also does not create any foul smell. Incinerators causes air pollution due to release of smoke and heat.



The Four R's

The four R's that we must adopt are **reduce**, **reuse**, **replace** and **recycle**.

Reduce

We must buy only those things that we need and can use. When we buy too many vegetables, some of them might rot and increase the amount of garbage in our house.

Here are some ways to reduce waste.



- ▶ Avoid buying things with lots of packaging material.
- ▶ Avoid buying throwaway products like plastic cups, glass and tins.
- ▶ Use a steel plate instead of a paper plate.

Reuse

Reusing means using those items again that otherwise would have been thrown away as waste.



Reusing saves money and puts waste to good use. Here are some ways of reusing items.

- ▶ We can reuse empty glasses or plastic jars for storing things such as pickles and dals.
- ▶ We can use biscuit and cheese tins for keeping things such as buttons, clips and needles.
- ▶ We can give the clothes and the toys we do not use to the needy.

Replace

We should replace non-biodegradable materials that we use in our daily life with biodegradable materials.

For example— We can replace plastic bags with cloth or jute bags.

Recycle

Recycling means converting used items into raw materials and then making new products from it.

We can use these new products again.

Example— Recycling paper.



A **kabadiwala** is a person who collects waste papers from our houses.

Other items we give to the **kabadiwala** are :

- ▶ In **metals** like cupboards, refrigerator and containers.
- ▶ In **plastics** like toys, bottles, bags and buckets.
- ▶ In **glass** like bulbs, jars and crockery.



Waste paper is brought to recycling factories. Here it is converted into pulp. Paper mills convert this pulp into paper again. Some factories make paper products like cards, paper flowers, paper plates, toilet paper, notebook paper and paper bags from recycled paper.





Advantages Of Recycling

1. Reduces amount of waste

Recycling reduces the amount of waste and solves the problem of waste disposal. It provides us raw material for making new articles again.

2. Saves natural resources

Fewer trees need to be cut to make newspapers and magazines. In the same way, a lesser amount of metals would be required when we start recycling metals.

REWIND



- ▶ Waste or garbage consists of materials that are no longer needed and are thrown away.
- ▶ There are two types of waste : bio-degradable and non-biodegradable.
- ▶ Wastes are disposed in three different ways : open dumping, sanitary land-filling and incineration.
- ▶ The four R's means that we should adopt : **Reduce**, **reuse**, **replace** and **recycle**.



A. Multiple choice questions.

Tick (✓) the correct option :

- Things that are of no use are called _____.
(a) resources (b) garbage
(c) minerals (d) none of these
- Converting used items into raw materials and then making new products from them, is called _____.
(a) recycling (b) replacing
(c) reducing (d) reusing



3. Waste paper can be converted into _____ .
- (a) glass (b) pulp
- (c) plastic (d) gum



B. Fill in the blanks :

- _____ waste stays unchanged into the soil for a long time.
- _____ are dug outside cities to dump garbage.
- We can replace plastic bags by _____ bags.

C. Answer the following questions :

- What is waste or garbage?

- Why is the sanitary land- filling method considered as a better method of disposing garbage?

- Differentiate between bio-degradable and non-biodegradable wastes.

- How can we help to reduce waste?

- Why do incinerators cause air pollution?

Extra Curriculum



- Write at least five lines on e-waste with the help of your teacher.



- _____
- _____
- _____
- _____
- _____



- Look at the pictures and name them.







- Draw a dustbin and write what all garbage we must throw in it.

-
-
-





Animals in Transport

In the past, men travelled only on foot. They carried their loads themselves. Later they learnt to domesticate animals. **The invention of the wheel brought a revolution in transport.** Carts fitted with wheels were used to carry people and goods from one place to another. These carts were pulled by animals.



Modes Of Transport Used In Past

Anita's grandfather had to travel all over India as part of his job. When he was young, **tongas, camel carts** and **bullock carts** were the main modes of transport in towns and cities. Surprisingly, you can still see them in small towns and villages.



Animals Used In Transport Today

Elephant

In the forest, elephants help to move heavy logs from one place to another.





Camel

The camel is known as the ship of the desert.

It is well adjusted to desert conditions. It can stay without water and food for several days. It can easily bear the heat of the desert.

Mounts

You must have seen people riding on horses, ponies and mules at hill stations. Have you ever tried riding on these animals? They are also used to carry goods over a distance. Such animals are called **mounts**.



In the mountains, animals such as donkeys, mules and yaks are used to carry loads. Dogs and reindeers are used to pull wheel-less carts called **sledges** in regions where the land is covered with ice and snow.



Special Occasions



You must have seen a bridegroom riding on a decorated horse in his marriage procession. Sometimes a luxury buggy is also used. A **buggy** is a four-wheeled luxury coach pulled by one or two horses. In some tourist places like sanctuary, people enjoy **elephant safari**.



As you know, brave children of our country are given bravery awards on the Republic Day. They also ride on beautifully decorated elephants to take part in the Republic Day Parade. You ride a camel or an elephant at some fairs and amusement parks as well.



Care Of Animals

Take proper care of your domestic animals.

- ▶ Give them proper food and clean drinking water.
- ▶ Give them proper shelter. A shelter protects them from heat, cold and rain. Clean their shelter regularly.
- ▶ Keep them clean by giving them a bath regularly.
- ▶ Do not force them to work beyond their capacity. Be kind to them.
- ▶ Do not beat animals when they refuse to work.
- ▶ When an animal falls sick, take it to veterinary.



REWIND



- ▶ Tongas, camel carts and bullock carts were the modes of main transport in the past.
- ▶ Animals used in transport today are mounts, elephants and camels.
- ▶ Elephant safari is used in special occasions.



A. Multiple choice questions.

Tick (✓) the correct option :

1. In the past, humans travelled on _____.
- | | |
|-----------|-----------|
| (a) cars | (b) foot |
| (c) boats | (d) buses |



2. Tongas are still a mode of transport in _____ .
- (a) cities (b) big towns
(c) villages (d) none of these
3. An _____ is used to transport heavy logs in the forest.
- (a) elephant (b) donkey
(c) camel (d) dog
4. When our pets fall sick we take them to a _____ doctor.
- (a) eye (b) veterinary
(c) specialist (d) none of these

B. Fill in the blanks :

1. The _____ is called the ship of the desert.
2. _____ are used to pull sledges in snowy regions.
3. A _____ is a four wheeled luxury coach pulled by horses.
4. When an animal falls sick, take it to a _____ doctor for treatment.

C. Write 'T' for true and 'F' for false :

1. Give animals proper food and clean drinking water.
2. Force animals to work beyond their capacity.
3. Do not beat animals when they refuse to work.
4. Keep animals clean by giving them a bath regularly.

D. Answer the following questions :

1. What are the main modes of transport in villages?
- _____
2. Why should we not force animals to work beyond their capacity?
- _____
3. How did the invention of the wheel bring about a revolution in transport?
- _____
4. Why is the camel called the 'ship of the desert'?
- _____



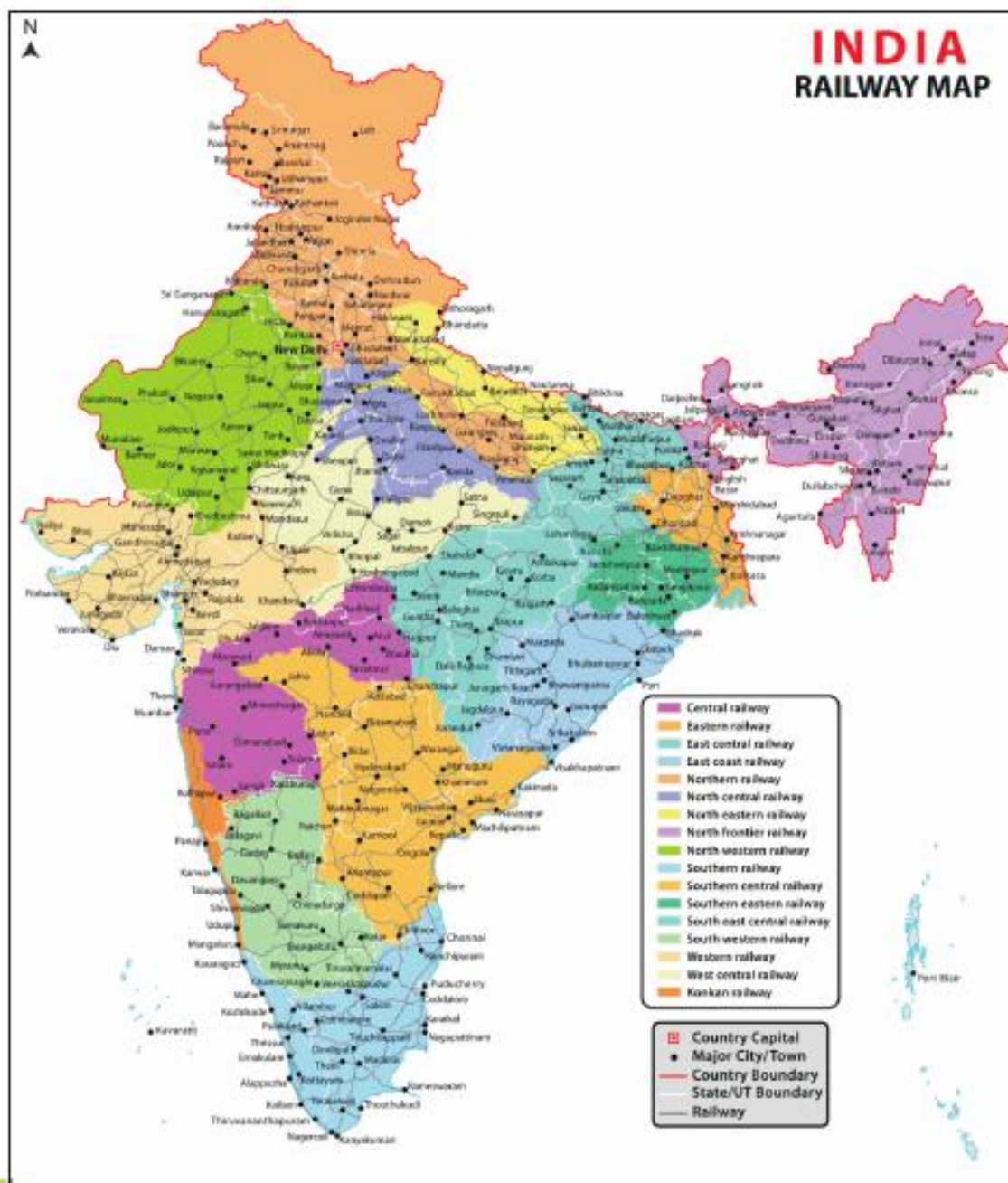
5. How should we take care of domestic and pet animals?

Extra Curriculum



- In the railway map of India, find out the three important junctions between : **Map Work 6**

- (a) Delhi and Mumbai
- (b) Delhi and Kolkata
- (c) Delhi and Thiruvananthapuram





Light, Shadows and Eclipses

The sun is a very important source of light. Without sunlight, there would be no plants or animals on Earth Surface. We cannot see colours in the dark. Our world would be a dead dark place without the light of the sun. People need light to see.



A human can see object when light falls on them.

The light of the sun is the main source of energy for all living beings. Besides the sun other sources provide light like- burning candle, electric bulb, etc.

Light is a type of energy that makes it possible for us to see the world around.



We need light to see. Light comes from different sources called sources of light. **For example-** sun, stars, torch etc.

All the objects we see around can be divided into two groups.

1. Luminous objects, 2. Non-luminous objects.

1. Luminous Objects

Objects which emit light of their own are called **Luminous** objects. **Examples-** sun, stars, candle, torch, electric bulb etc. There are some animals that create their own light, such as- jellyfish, fireflies etc.



2. Non- luminous Objects

Objects which do not emit light of their own. They reflect when the light falls on them are called Non-luminous objects. **Example-** moon, earth, plants, table, book etc.



Transparent, Translucent And Opaque



Objects are further divided into three groups on the basis of the amount of light that can pass through them.

1. Transparent object, 2. Translucent object, 3. Opaque object



1. Transparent Object

Some objects let light to pass through them completely are known as transparent objects. Glass, water, and air are some examples of transparent objects.

2. Translucent Object

Some objects let some light to pass through them are known as translucent objects. Wax paper, butter paper, vegetable oil are some examples of translucent objects.



Opaque Object – Some objects block light that falls on them are known as opaque objects. Wood, brick, stone, book, animals, plants are some examples of opaque objects.



The Speed Of Light

Light moves at the fastest known speed in the universe. Nothing moves faster than the speed of light. Light from the sun takes about 8 minutes and 20 seconds to reach the earth surface.





Rainbow is a natural phenomenon. It is a semi-circle consists of seven colours. These colours are violet, indigo, blue, green, yellow, orange and red. Rainbow is usually formed before and after rainfall.



A telescope is an instrument used to see objects that are far away from us. People generally use telescopes to see objects like planets, stars, comets, moon and satellites etc. Sometimes telescopes are used to view scenery, sports, wild life and nature.



Some uses of light can be listed as:

1. It helps to see the objects around us.
2. It is used by the plants to make their food and grow.
3. It is used in many electronic appliances.
4. It is used in power satellites.
5. It keeps the earth warm.
6. The light from the sun is used in solar panel which are used in domestic appliances, such as- solar cooker.
7. Sunlight helps in keeping nature clean. Due to drying, there is no growth of germs.
8. Sunlight is a source of energy as well as a source of vitamin D for human beings.



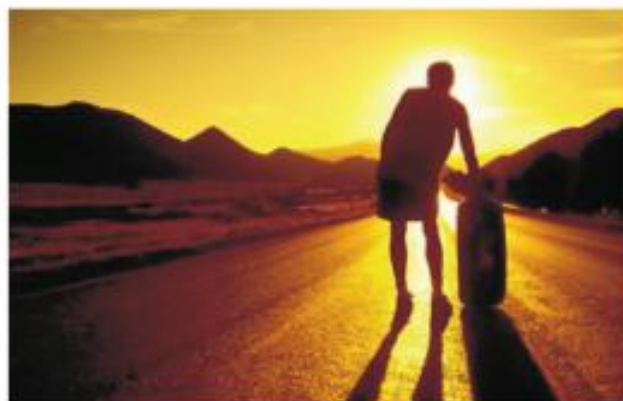


Shadows

A shadow is formed when an object blocks a light source. Generally, shadows appear black in colour. The size of shadow depends on the position of the object from the source of light.

Shadow in Morning and Evening

Shadows are long in the morning and evening because the sun rays are slanting on the body in the morning and evening.



Shadow at Noon

At noon, the sun rays fall directly over the head or on the body. So, the shadow is very short at noon.



Eclipse

An eclipse occurs when one heavenly body such as a moon or planet moves into the shadow of other heavenly body.

Types of Eclipse

There are only two types of eclipses— Solar eclipse and Lunar eclipse.

Solar Eclipse

A solar eclipse occurs when the moon comes between the earth and the sun. This makes the moon fully or partially cover the sun. It looks like the sun has gone dark. It happens only on a new moon day.



Lunar Eclipse

A lunar eclipse occurs when the earth comes in between the sun and the moon. A lunar eclipse always occurs only on the night of a full moon.



REWIND



- ▶ We cannot see objects in the dark.
- ▶ Sun light is the main source of energy for all living beings.
- ▶ Objects which emit light of their own are called luminous objects.
- ▶ Objects which do not emit light of their own and reflect when the light falls on them are called non-luminous objects.
- ▶ Objects are of 3 types on the basis of amount of light that can pass through them - Transparent, Translucent, Opaque.
- ▶ Rainbow is a semi- circle of seven colours.
- ▶ Telescope is used to see distant objects.
- ▶ Shadow in the morning and evening are long and at noon it is short.
- ▶ Eclipse are of two types- solar eclipse and lunar eclipse.



A. Multiple choice questions.

Tick (✓) the correct option :

1. An object which is a source of light of its own _____.



- | | | | |
|------------------|---|--------------|---|
| (a) non-luminous |  | (b) luminous |  |
| (c) opaque |  | (d) shadow |  |
- Which one of the following is the natural source of light?

(a) tube light		(b) torch	
(c) moon		(d) sun	
 - A material through which light can pass completely.

(a) opaque		(b) transparent	
(c) translucent		(d) filter	
 - _____ eclipse happens when the moon comes in between the earth and the sun.

(a) solar		(b) both (a) and (b)	
(c) lunar		(d) none of the above	
 - Telescope is used to _____.

(a) measure		(b) watch television	
(c) see heavenly bodies		(d) none of the above	

B. Fill in the blanks :

- Our world would be a dead dark place without the _____.
- Some objects let some light to pass through them are known as _____ objects.
- Light from the sun takes _____ to reach the earth surface.
- Rainbow is usually formed _____ and _____ rainfall.
- Sunlight is a source of _____ for human.
- Solar eclipse happens only a _____.

C. Give two example for each :

- | | | |
|-------------------------|-------|-------|
| 1. Luminous objects | _____ | _____ |
| 2. Non-luminous objects | _____ | _____ |
| 3. Transparent objects | _____ | _____ |
| 4. Translucent objects | _____ | _____ |
| 5. Opaque objects | _____ | _____ |
| 6. Types of eclipse | _____ | _____ |



D. Write 'T' for true and 'F' for false :

1. Sun light is the main source of energy for all living beings.
2. Jellyfish do not create their own light.
3. Some objects block light that falls on them are known as translucent objects
4. Nothing moves faster than the speed of light.
5. Telescope helps us to see objects near by us.



E. Answer the following questions in short :

1. Define the term of light.

2. Give two examples of animals that emit their own light.

3. What are the seven colours of a rainbow?

4. What are opaque objects? Give two examples.

5. When do we see a rainbow?

F. Answer the following questions in detail :

1. Define luminous and Non-luminous objects with examples .
2. What is a telescope?
3. Give four uses of light.
4. What is a lunar eclipse? Draw a diagram to show the formation of the lunar eclipse.
5. Draw a neat and clean diagram of solar eclipse.

Extra Curriculum



- How to make a homemade solar eclipse. Ask your teacher and make a list of things which you need.

Communication



Put the objects in the correct columns.



1



2



3



4



5



6



7



8

Non-luminous Objects

Luminous Objects

**A. Multiple choice question.**

Tick (✓) the correct option:

1. We should brush our _____.
(a) teeth (b) gums
(c) hair (d) tongue
2. Exercise keeps our body _____.
(a) fit (b) healthy
(c) both (a) and (b) (d) none of these
3. When a person gets pain, use an _____.
(a) antiseptic (b) injection
(c) ice-pack (d) all of these
4. India Gate was built to pay _____ to the martyrs of the first world war.
(a) homage (b) honour
(c) respect (d) none of these
5. We eat the leaves of plants such a _____.
(a) cauliflower (b) carrot
(c) cabbage (d) radish

B. Fill in the blanks :

1. Drinking polluted water causes diseases like _____ and _____.
2. _____ are dug outside cities to dump garbage.
3. The wheel is of Tiranga has 24 _____.
4. Leaves of plants make food by process called _____.
5. The full form of ASI _____.

C. Answer the following questions :

1. How should we take care of our eyes?
2. How can you reduce pollution?
3. Which organs of the body help us in respiration?
4. Why are historical monuments important to us?
5. What is photosynthesis ? Explain the process?



**A. Multiple choice question.****Tick (✓) the correct option:**

1. Harmful substance in the environment are called _____.
(a) pollutants ☐ (b) wastes ☐
(c) garbage ☐ (d) pollution ☐
2. The Gir National Park in Gujarat is home to _____.
(a) tigers ☐ (b) lions ☐
(c) rhinoceros ☐ (d) leopards ☐
3. Shaking or trembling of the earth's surface is called an _____.
(a) cyclone ☐ (b) landslide ☐
(c) earthquake ☐ (d) drought ☐
4. A renewable source of energy is _____.
(a) petrol ☐ (b) coal ☐
(c) sun ☐ (d) moon ☐
5. A material through which light can pass completely is _____.
(a) opaque ☐ (b) transparent ☐
(b) translucent ☐ (d) filter ☐

B. Fill in the blanks.

1. _____ treats people who are ill or injured.
2. A patch of forest land dedicated to deity is _____.
3. The full form CNG is _____.
4. _____ is a long period of time when there is no rain.
5. Rhinoceros is being killed for its _____.

C. Answer the following questions.

1. Write the work done by these people: (a) Doctor, (b) Tailor, (c) Cobbler
2. What are fuels? Name any two fuels.
3. Why are wood houses made in hilly, forested areas?
4. What is Project Tiger?
5. What is lunar eclipse? Draw a neat diagram to show the formation of lunar eclipse.

